

Exercise Week 2

Today's exercises

Exercise 2-1: Drive DC motor using transistor

Exercise 2-2: Drive DC motor using motor driver IC

Exercise 2-3: RC servomotor

Exercise 2-4: Infrared distance sensor

Exercise 2-5: Digital low-pass filter & serial plotter

Exercise 2-6: Ultrasonic distance sensor

Exercise 2-7: Micro switch

Show the result of each exercise to an instructor or TA, every time you complete an exercise

Exercise 2-1

1. Make a circuit on a breadboard to drive a **motor** using a **MOSFET** (2SK4017) and a **diode** (1N5817)
2. Make a program to send a **PWM** signal of **period** 0.02 and change **duty cycle** at every 1 sec as $0 \rightarrow 0.1 \rightarrow 0.2 \rightarrow 0.3 \rightarrow 0.4 \rightarrow 0.5 \rightarrow 0.4 \rightarrow 0.3 \rightarrow 0.2 \rightarrow 0.1 \rightarrow 0$
3. Then, show the result and **explain**
 - **purpose of the MOSFET**
 - **purpose of the diode**
 - **purpose of resistors in the circuit**to an instructor or a TA

➤ See NEXT page for the procedure

Procedure

Step 1: See P. 5–11 to learn about the MOSFET

Step 2: Choose a pin of microcontroller used for PWM (P. 12)

Step 3: Make a circuit on a breadboard (P. 13-15)

Step 4: Make a new project (P. 7-10 of “[platformIO.pdf](#)”)

Step 5: Write a code in main.cpp (P. 16-17)

- Use [Mbed API “PWMOut”](#) to generate PWM signals

- Use for-loop to change the “duty cycle”

Step 6: Connect a microcontroller to your PC

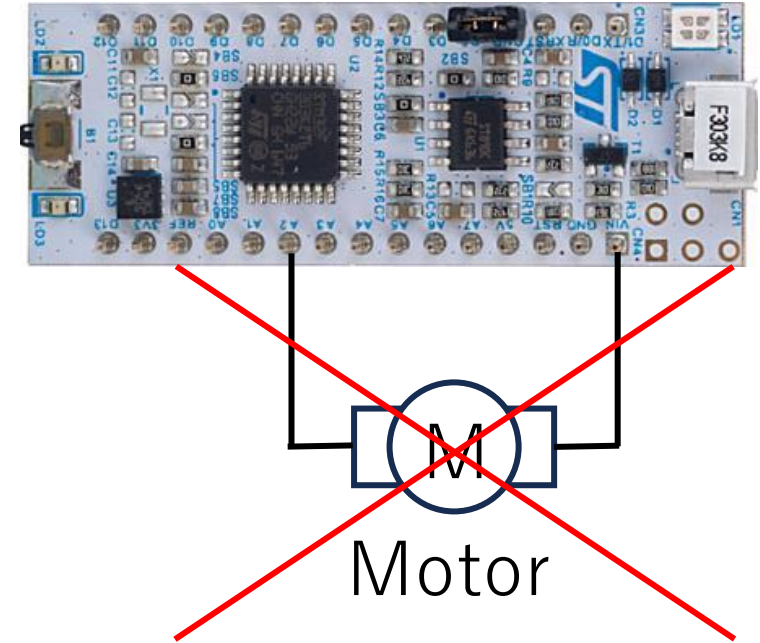
Step 7: Build & run the project (P. 14 of “[platformIO.pdf](#)”)

- See P. 16-18 of “[platformIO.pdf](#)”, if upload fails

Ask an instructor or TA questions, if you don't understand

Why do you need a transistor ?

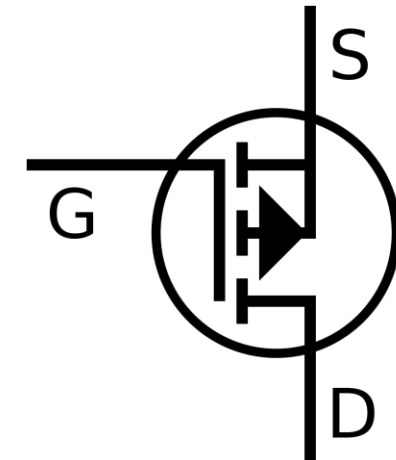
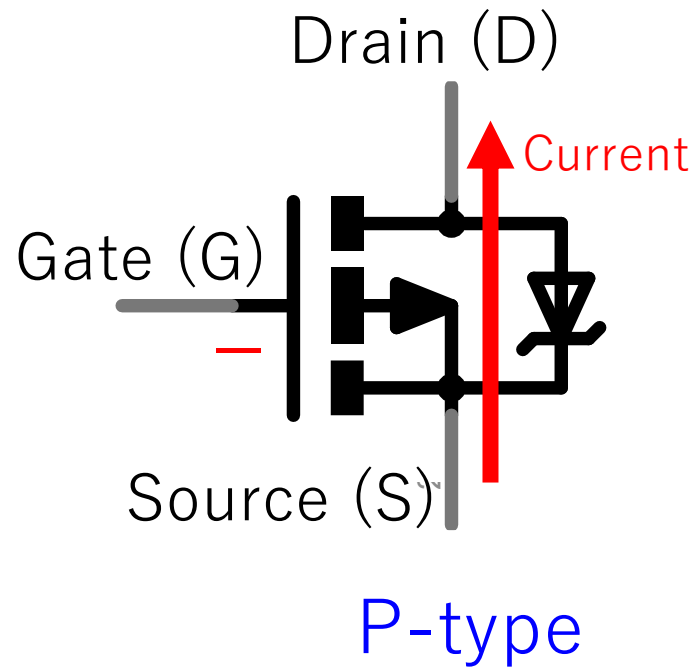
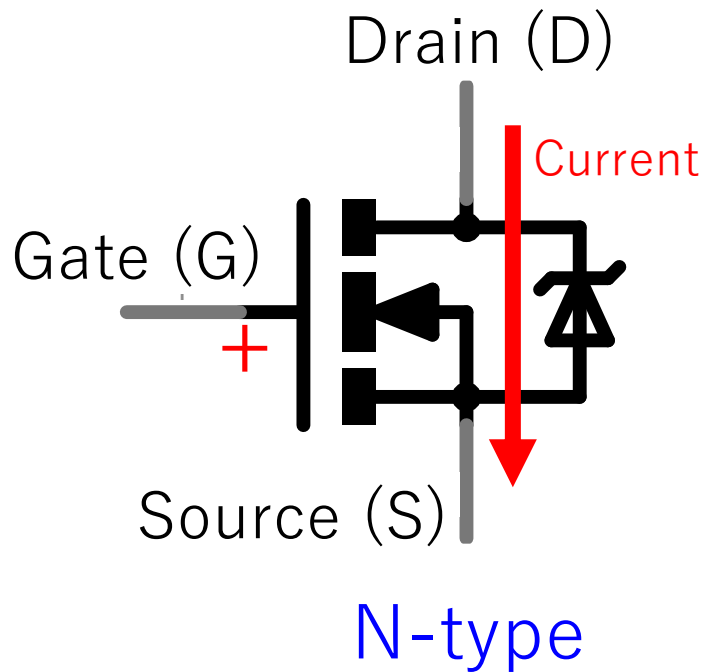
- Currents from microcontrollers are too small to drive a motor
 - Max 25mA/pin
 - Max 80mA total
- in the case of NUCLEO-F303K8
- External power source (e.g., battery) is required
- Transistor is used to switch ON and OFF the external power supply
- MOSFET (P. 6) is a popular type of transistors



Microcontroller will be broken, if you try to drive a motor directly

MOSFET (Metal Oxide Semiconductor Field Effect Transistor)

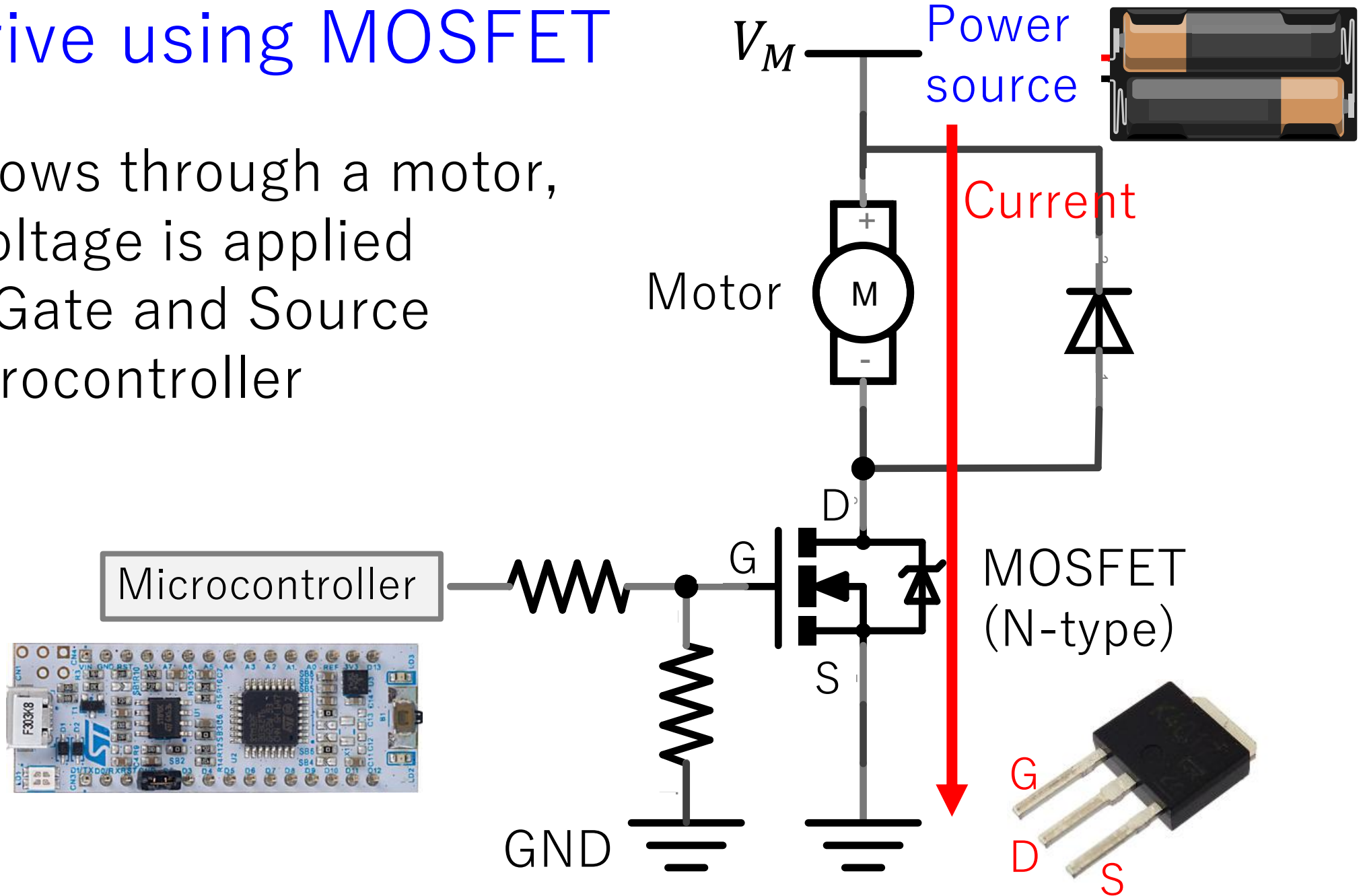
Current flows between Drain and Source, when a voltage is applied between Gate (G) and Source (S)



Another type of symbol (P-type)

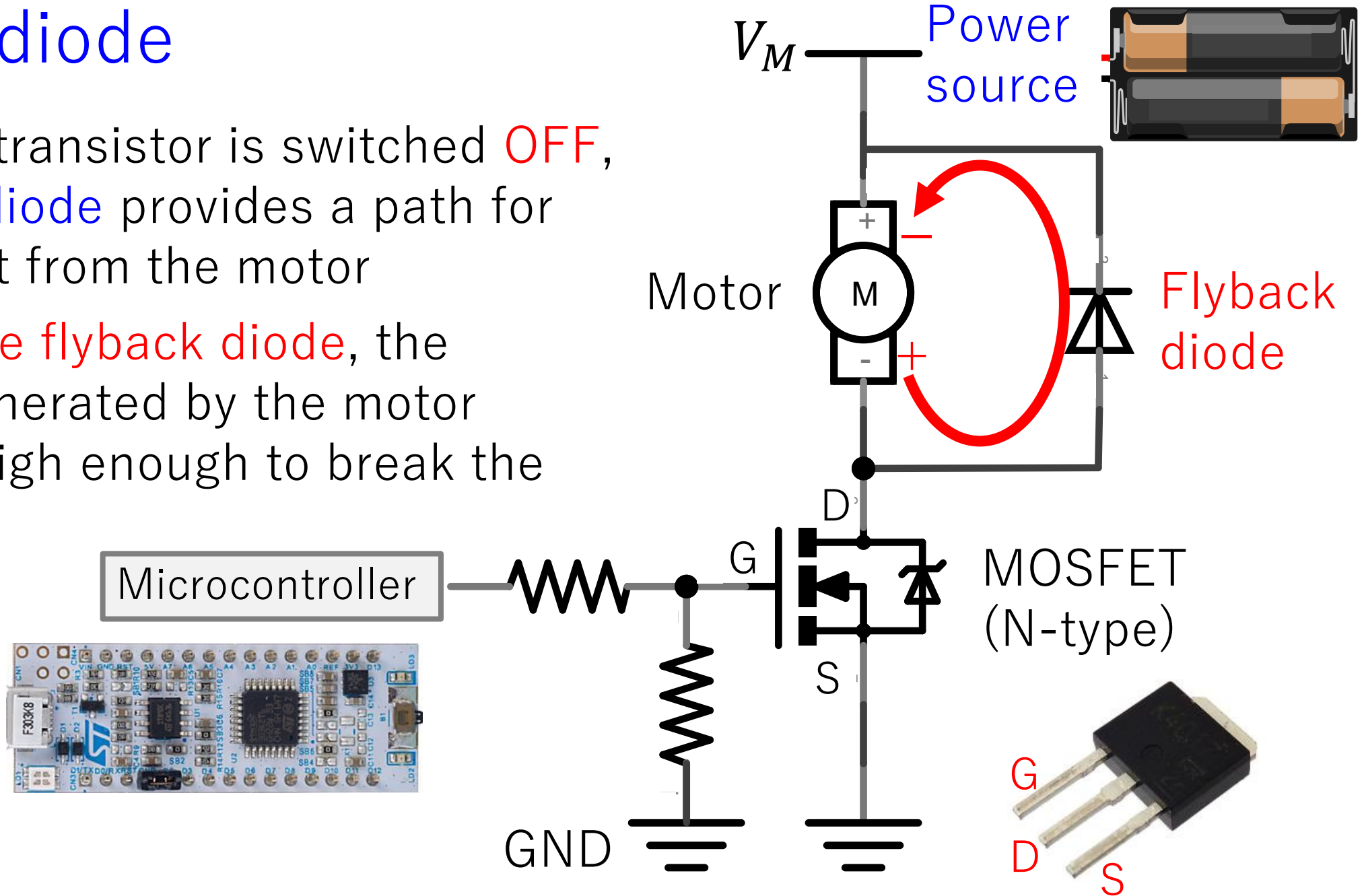
Motor drive using MOSFET

Current flows through a motor, when a voltage is applied between Gate and Source using microcontroller



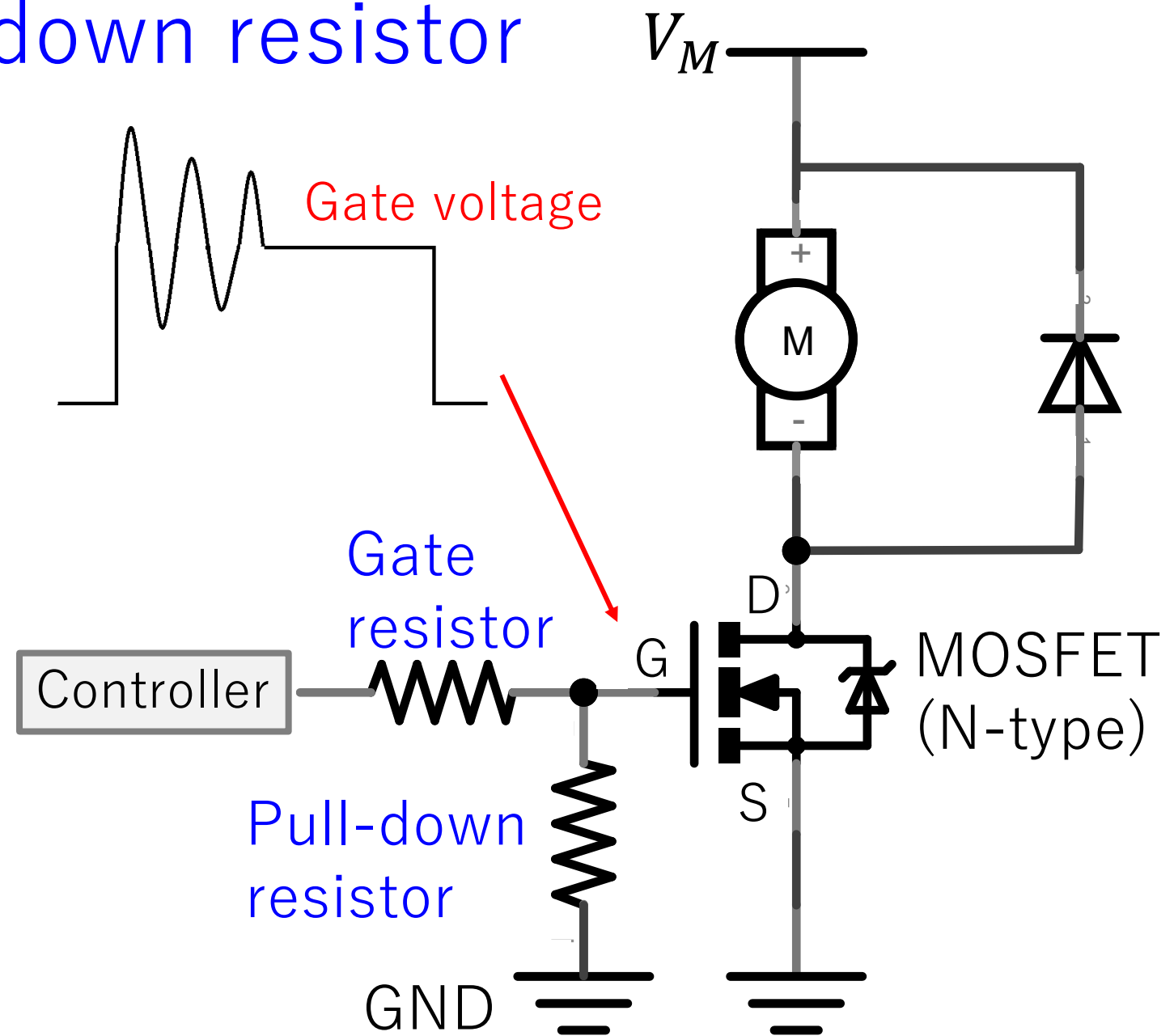
Flyback diode

- When the transistor is switched **OFF**, a **flyback diode** provides a path for the current from the motor
- **Without the flyback diode**, the voltage generated by the motor could be high enough to break the transistor



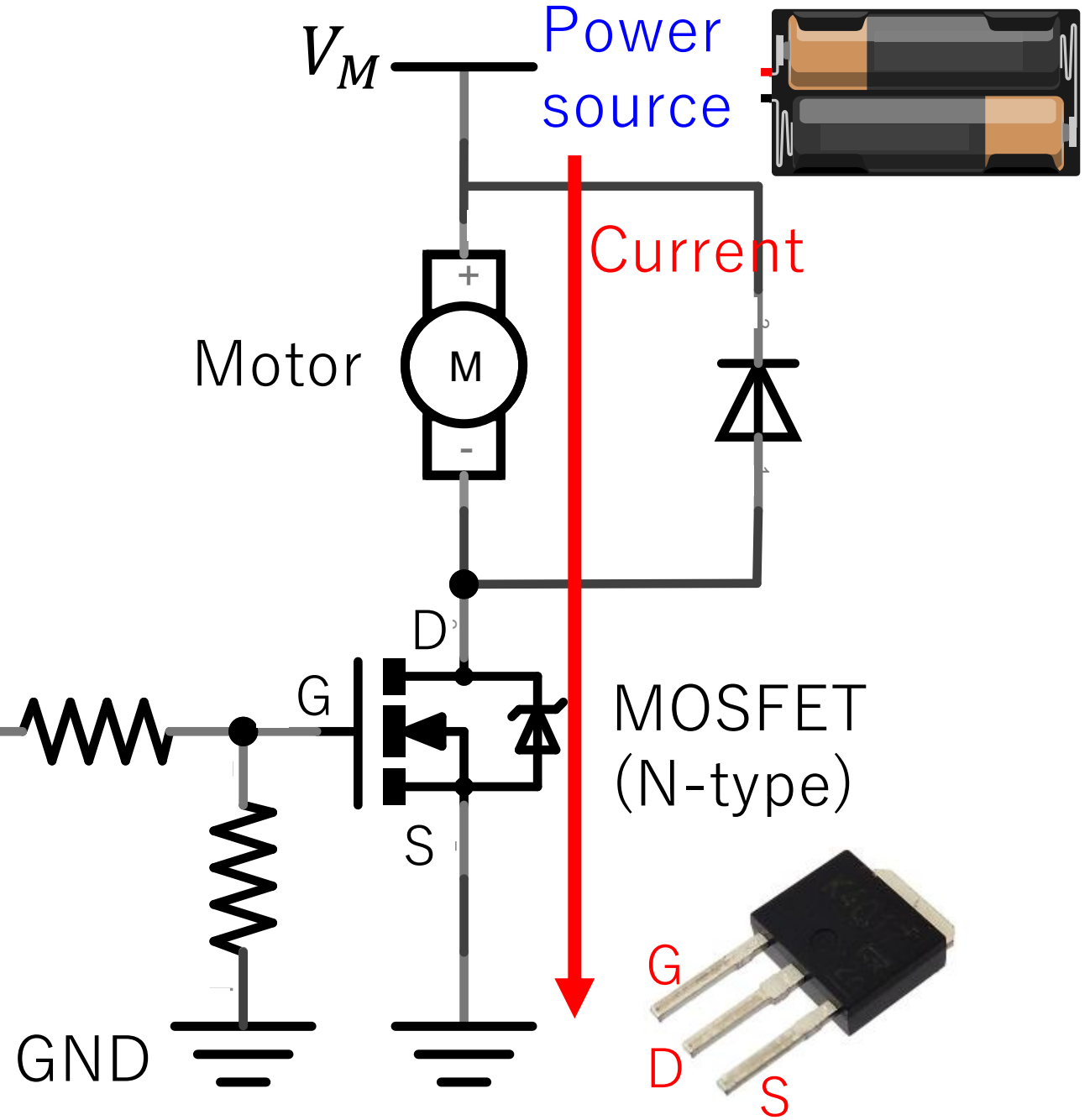
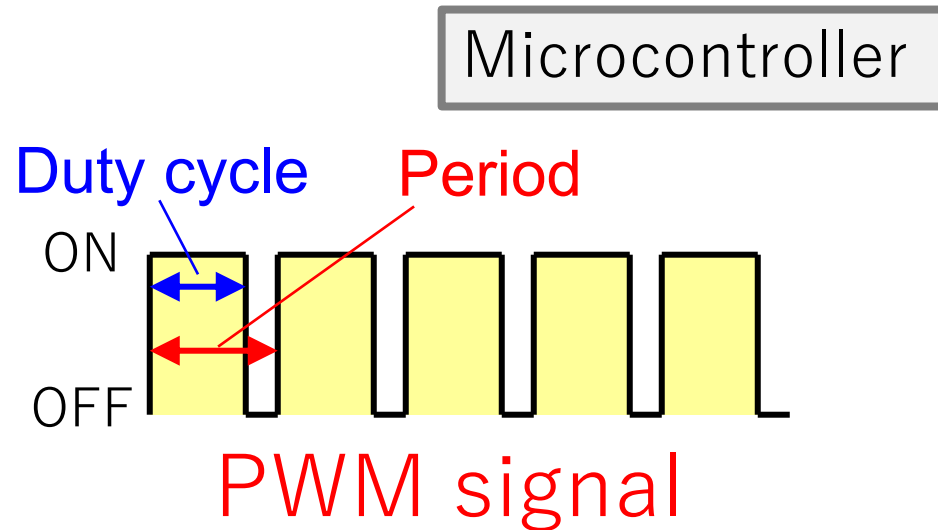
Gate resistor & pull-down resistor

- Gate resistors prevent **oscillations** of the gate voltage and **excessive currents** through microcontrollers
- Pull-down resistors turn OFF MOSFET (by applying 0V), when there is no signal from microcontrollers



Changing motor speed

Motor speed changes by changing the voltage between Gate and Source
→ Change the duty cycle of **PWM signal** (P. 11) from microcontroller

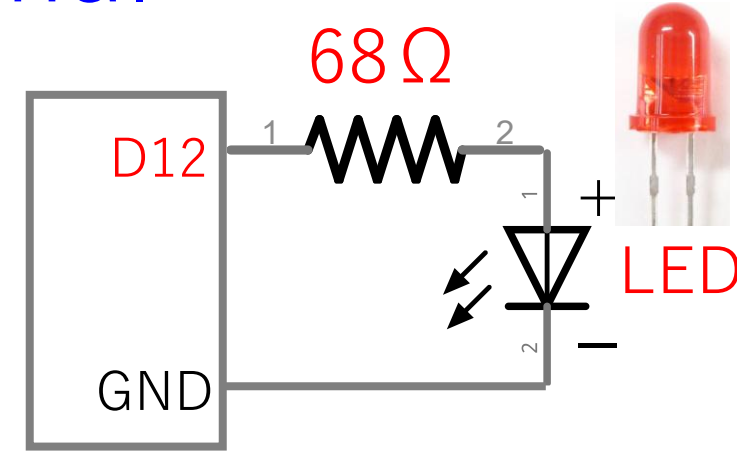


PWM (Pulse-width modulation) signal

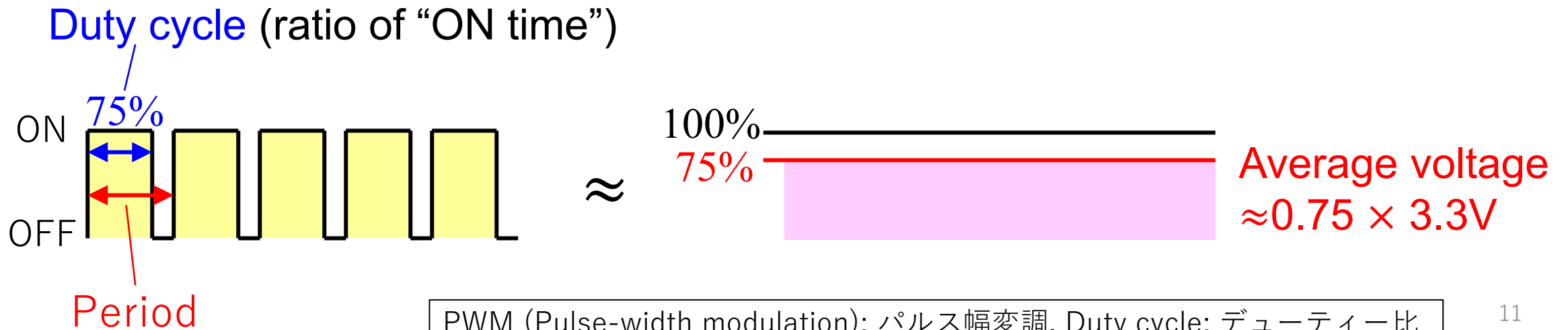
Voltage from a microcontroller pin is fixed ($\approx 3.3\text{V}$)
→ How do you change the brightness of a LED ?



Switch ON&OFF voltage supply for a **short period**
→ **Duty Cycle** $\times 3.3\text{V} \approx \text{Average Voltage}$

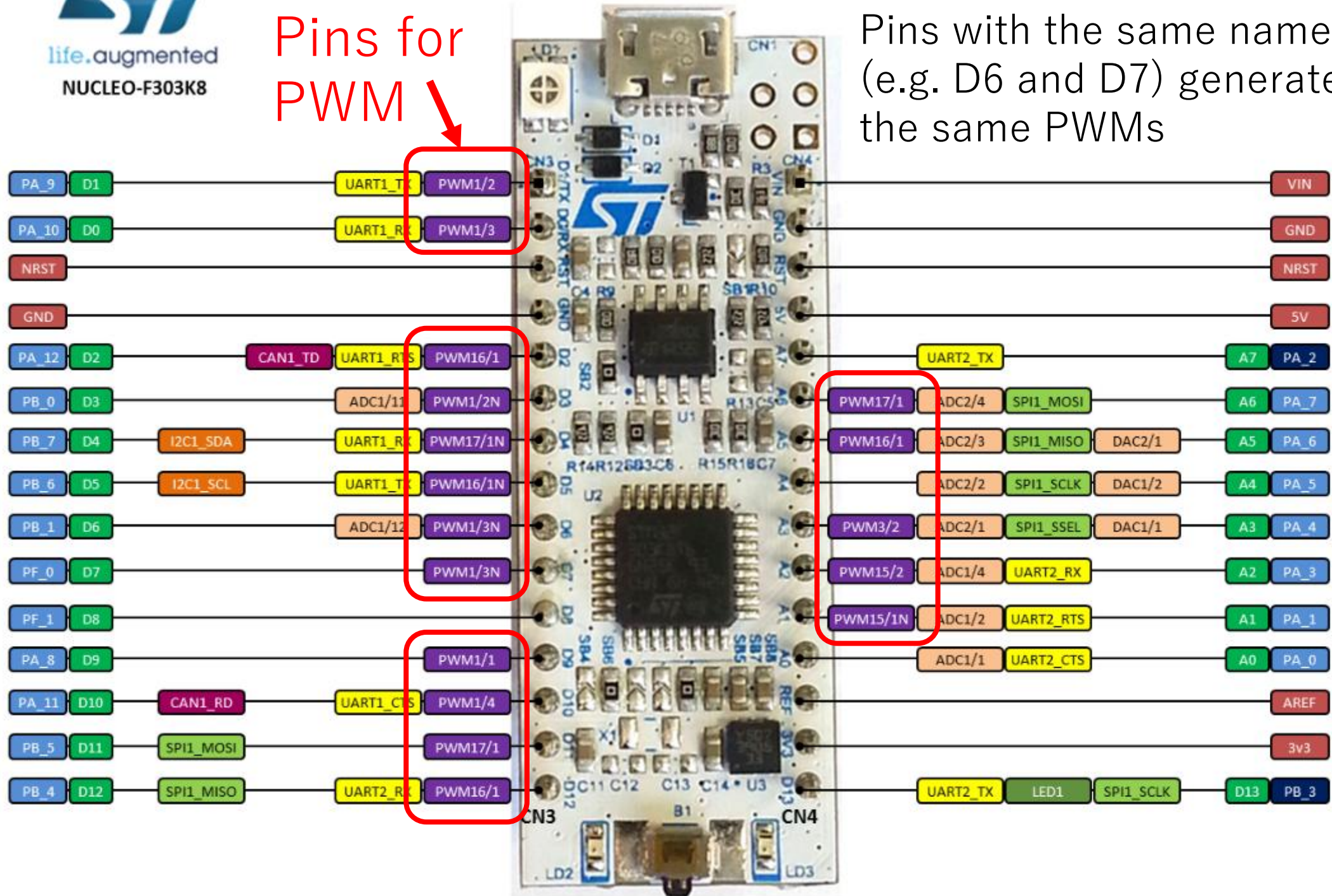


Example of PWM signal



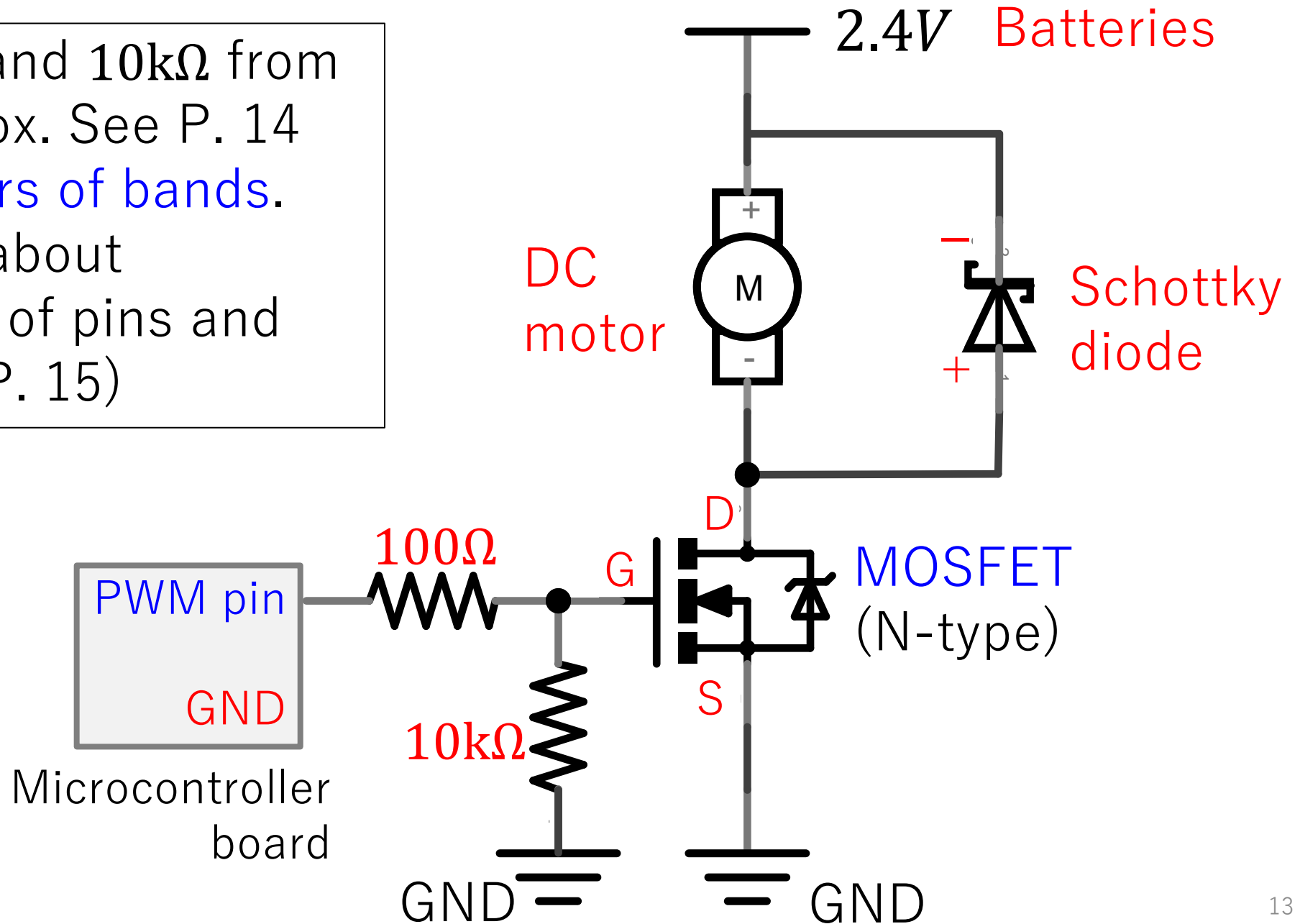
Pins for
PWM

Pins with the same name
(e.g. D6 and D7) generate
the same PWMs



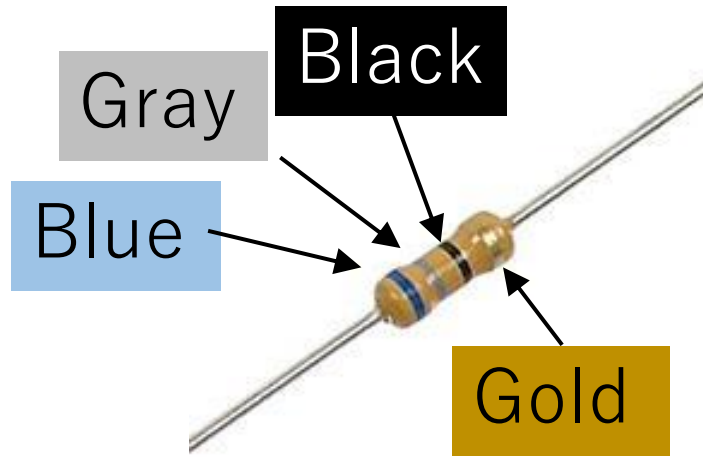
Step 3: Make a circuit on a breadboard

- Find 100Ω and $10k\Omega$ from your part box. See P. 14 for the **colors of bands**.
- Be careful about differences of pins and polarities (P. 15)



Resistors

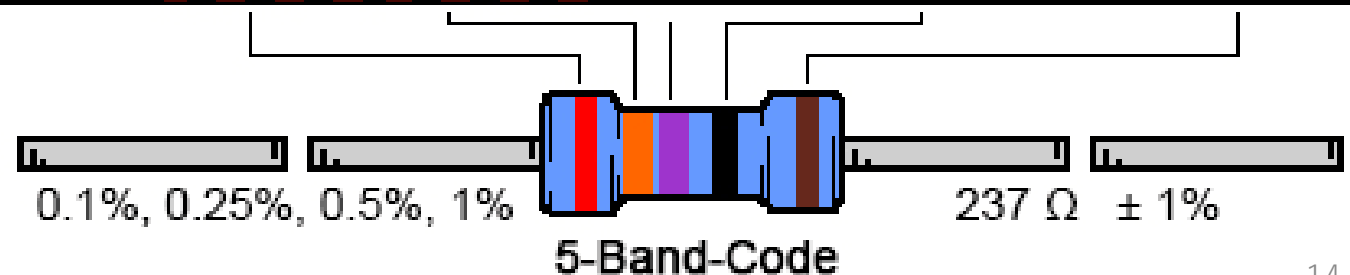
Example: 68 Ω



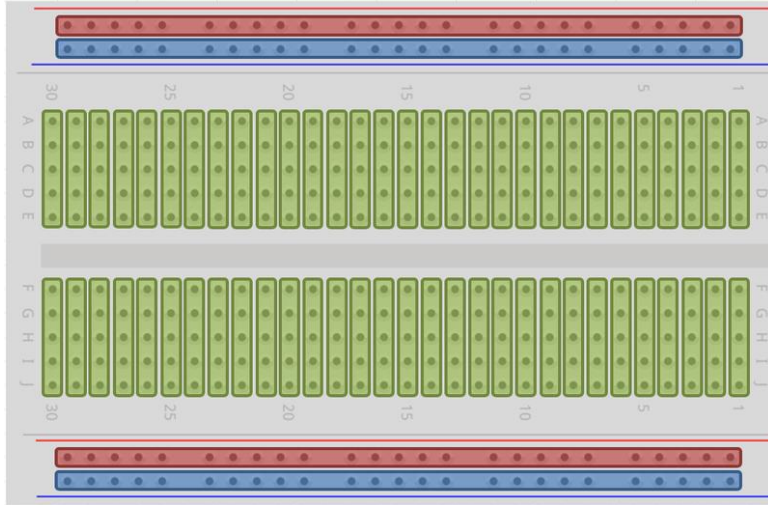
Blue Gray Black
 68×10^0
 Tolerance: ± 5 (%)
 許容差



COLOR	1 ST BAND	2 ND BAND	3 RD BAND	MULTIPLIER	TOLERANCE
Black	0	0	0	1Ω	
Brown	1	1	1	10Ω	$\pm 1\%$ (F)
Red	2	2	2	100Ω	$\pm 2\%$ (G)
Orange	3	3	3	1KΩ	
Yellow	4	4	4	10KΩ	
Green	5	5	5	100KΩ	$\pm 0.5\%$ (D)
Blue	6	6	6	1MΩ	$\pm 0.25\%$ (C)
Violet	7	7	7	10MΩ	$\pm 0.10\%$ (B)
Grey	8	8	8	100MΩ	$\pm 0.05\%$
White	9	9	9	1GΩ	
Gold				0.1Ω	$\pm 5\%$ (J)
Silver				0.01Ω	$\pm 10\%$ (K)



Breadboard



Connection of holes

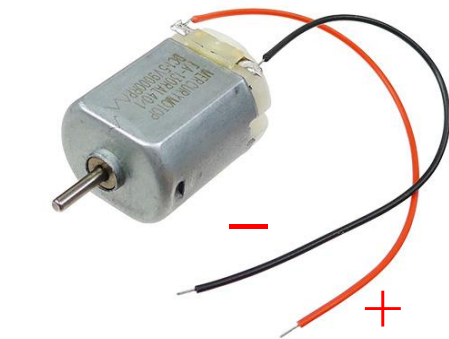
<https://nobita-rx7.hatenablog.com/entry/27352290>



Schottky
diode

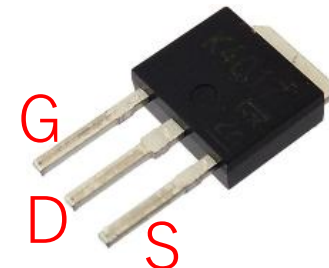
1N5817

[Datasheet](#)

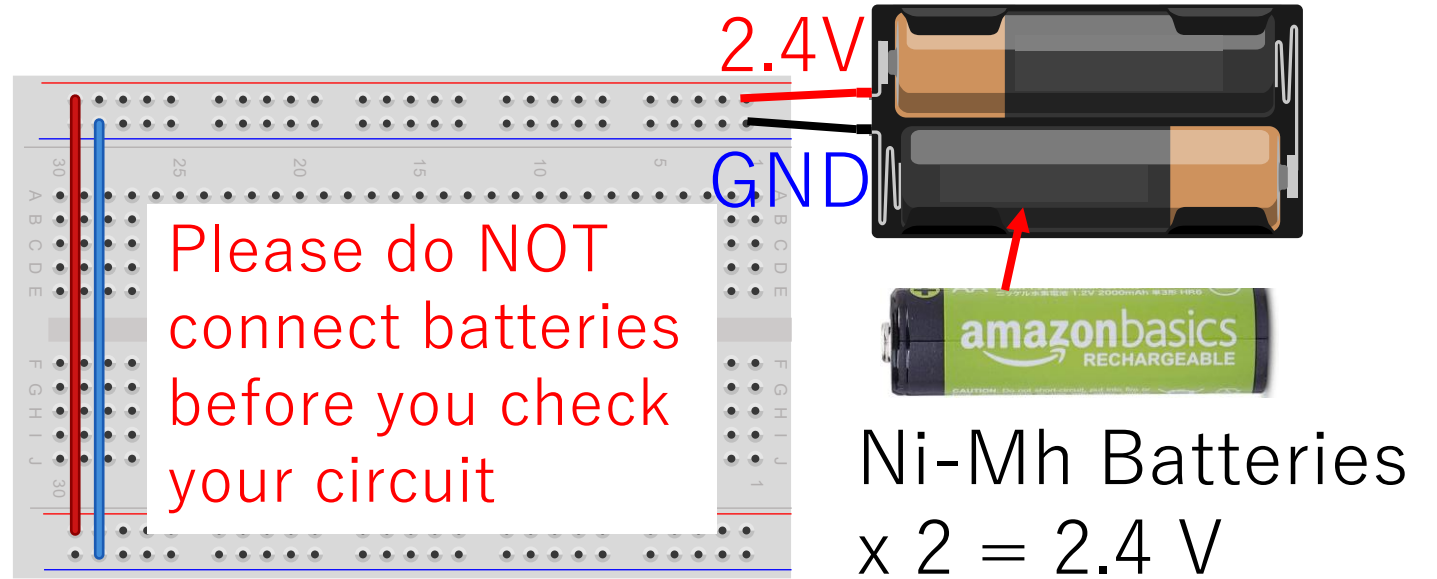
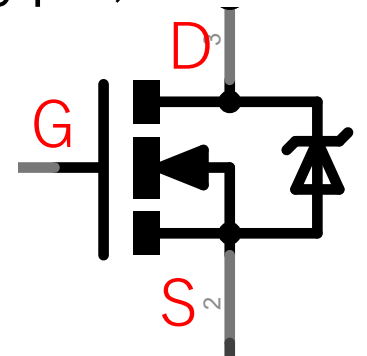


FA-130RA-2270

MOSFET (N-type)



2SK4017



Ni-Mh Batteries
 $\times 2 = 2.4 \text{ V}$

Mbed API: “PWMOut”

[Click for Documentation](#)

Initialization:

```
PWMOut object_name(Pin_name);
```

Class

Functions of PWMOut :

- period()
- write()
- read(), etc



Example:

Set the PWM period as 0.02 sec
And the duty cycle as 3%

```
object_name.period(0.02f)  
object_name.write(0.03f)
```

object_name = 0.03f

- The default period is 0.020s, and the default duty cycle is 0.
- “0.02f” means 0.02 of the float type. Without “f”, it means the double.

For-loop

Syntax:

```
for (Initialization; condition; update) {  
    // code to be executed  
}
```

- Example 1 (integer):

```
for (int i = 0; i < 10; i++) {  
    // code to be executed  
}
```

- Example 2 (float):

```
for (float p = 0.0f; p <= 1.0f; p+=0.1f) {  
    // code to be executed  
}
```

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Exercise 2-3: RC servomotor

Exercise 2-4: Infrared distance sensor

Exercise 2-5: Digital low-pass filter & serial plotter

Exercise 2-6: Ultrasonic distance sensor

Exercise 2-7: Micro switch

Show the result of each exercise to an instructor or TA, every time you complete an exercise

Exercise 2-2

1. Make a circuit on a breadboard to drive a **motor** using the **motor driver IC** (AE-DRV8835-S)
2. Make a program to send a **PWM** signal of **period** 0.02s and change the **rotation direction & duty cycle** every 1 sec as:
Forward & 0.2 → **Brake** → **Backward & 0.2** → **Coast**
3. Then, show the result and **explain**
 - **purpose of H-bridge circuits in the IC**
 - **how to change the rotational speed of the motor**
 - **how to use one more motor**to an instructor or a TA

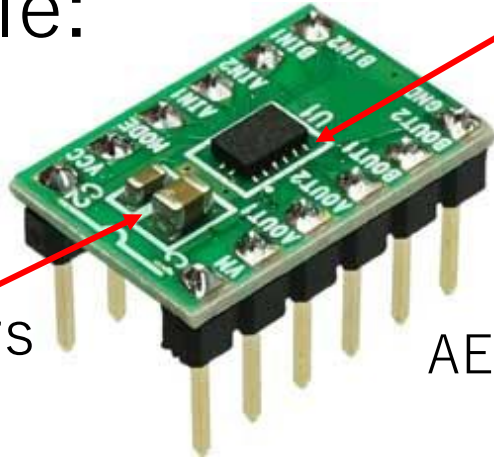
➤ See NEXT page for the procedure

Procedure

- Step 1: See P. 21-24 to learn about the motor driver IC
- Step 2: Select two pins of microcontroller used for PWM (P. 12)
- Step 3: Make a circuit on a breadboard (P. 25)
- Step 4: Make a new project of Mbed OS (P. 16-18)
- Step 5: Write a code in main.cpp (P. 26)
- Step 6: Connect a microcontroller to your PC
- Step 7: Build & run the project (P. 14 of “[platformIO.pdf](#)”)
 - See P. 16-18 of “[platformIO.pdf](#)”, if upload fails

Motor driver IC

Example:



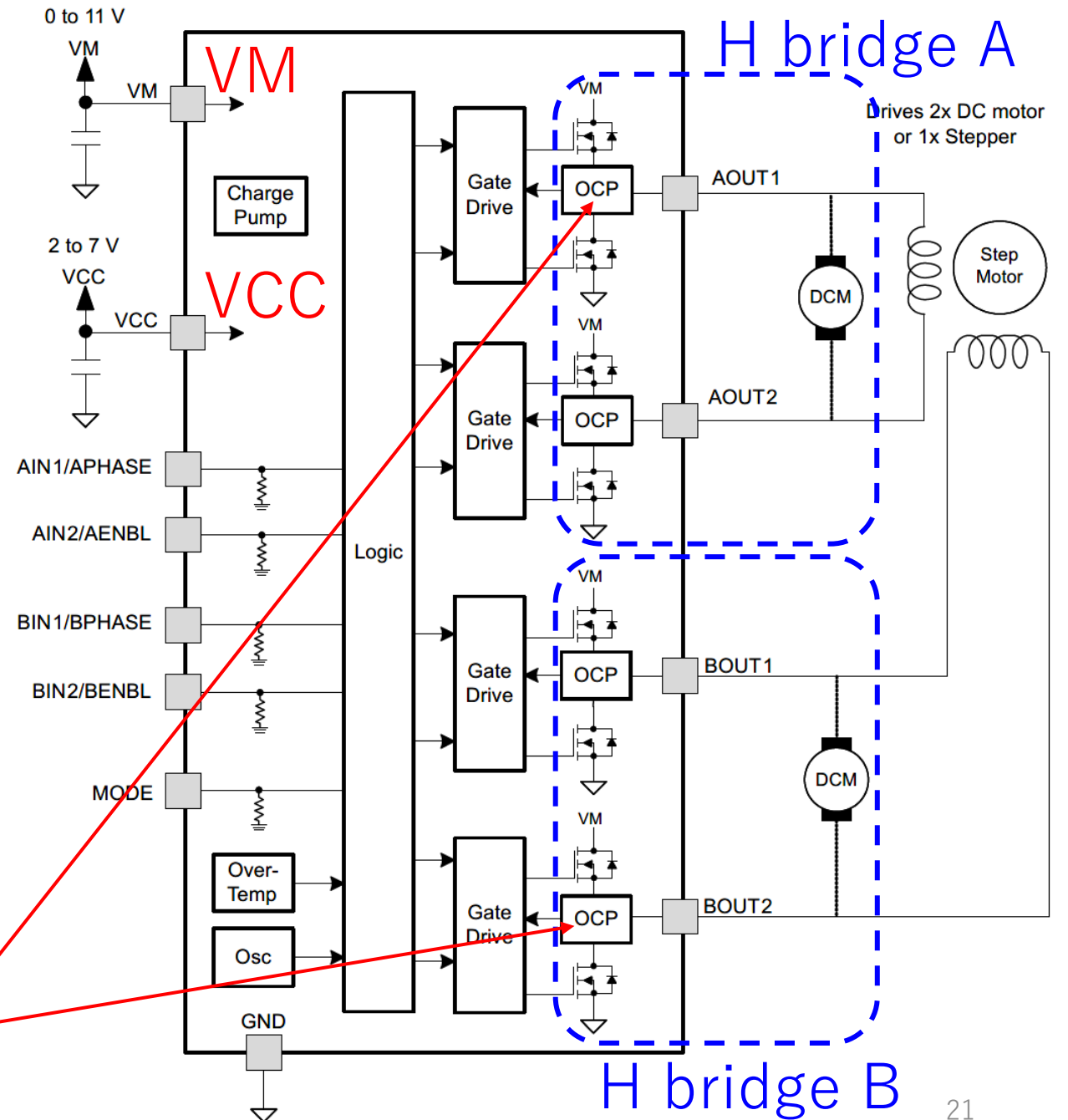
DRV8835
[Datasheet](#)

AE-DRV8835-S
[Manual](#)

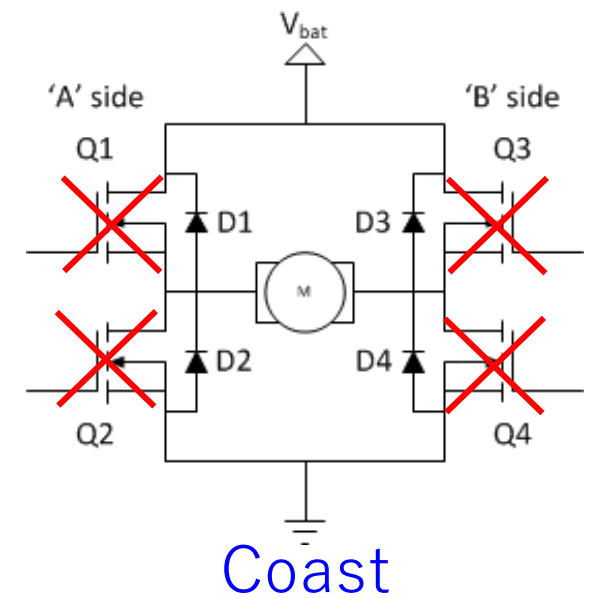
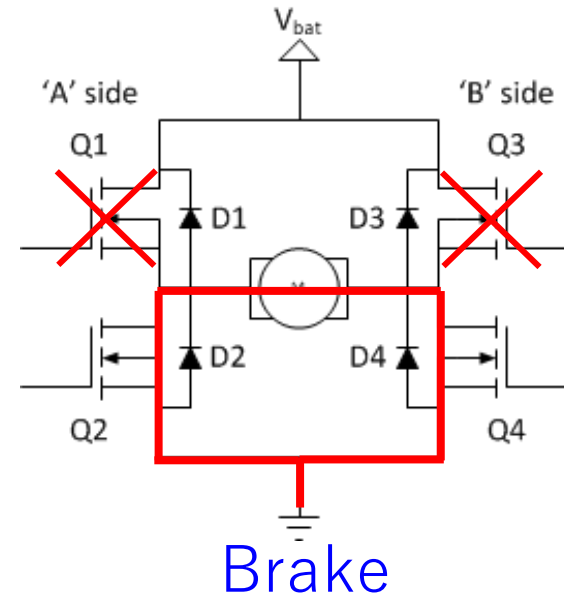
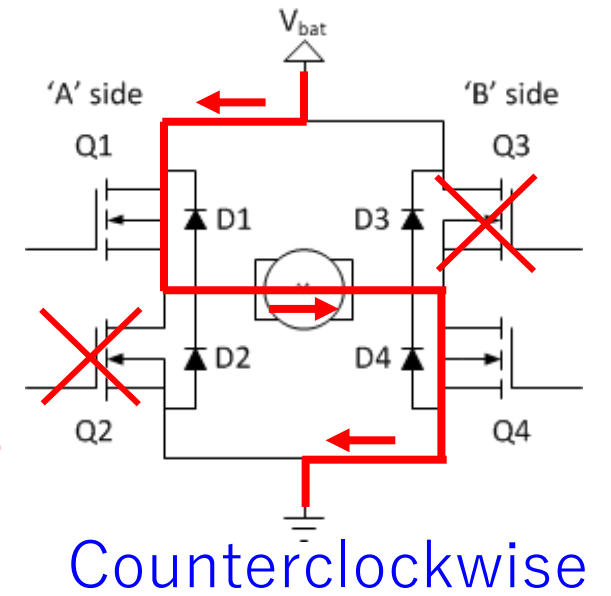
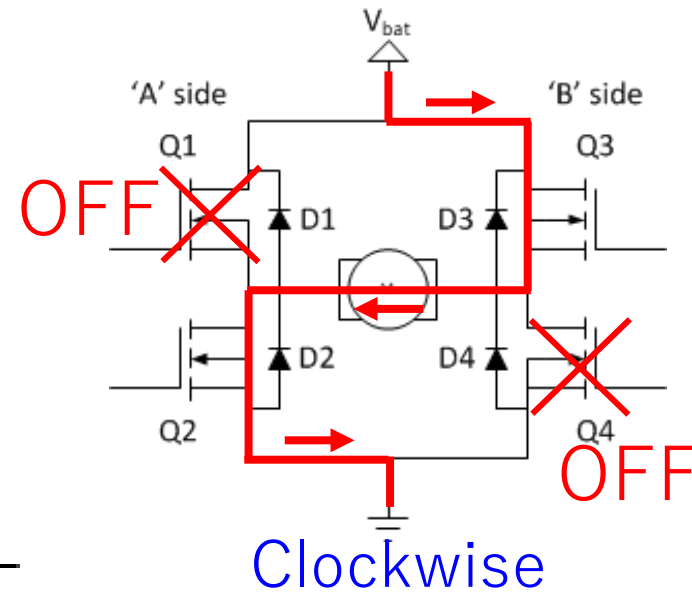
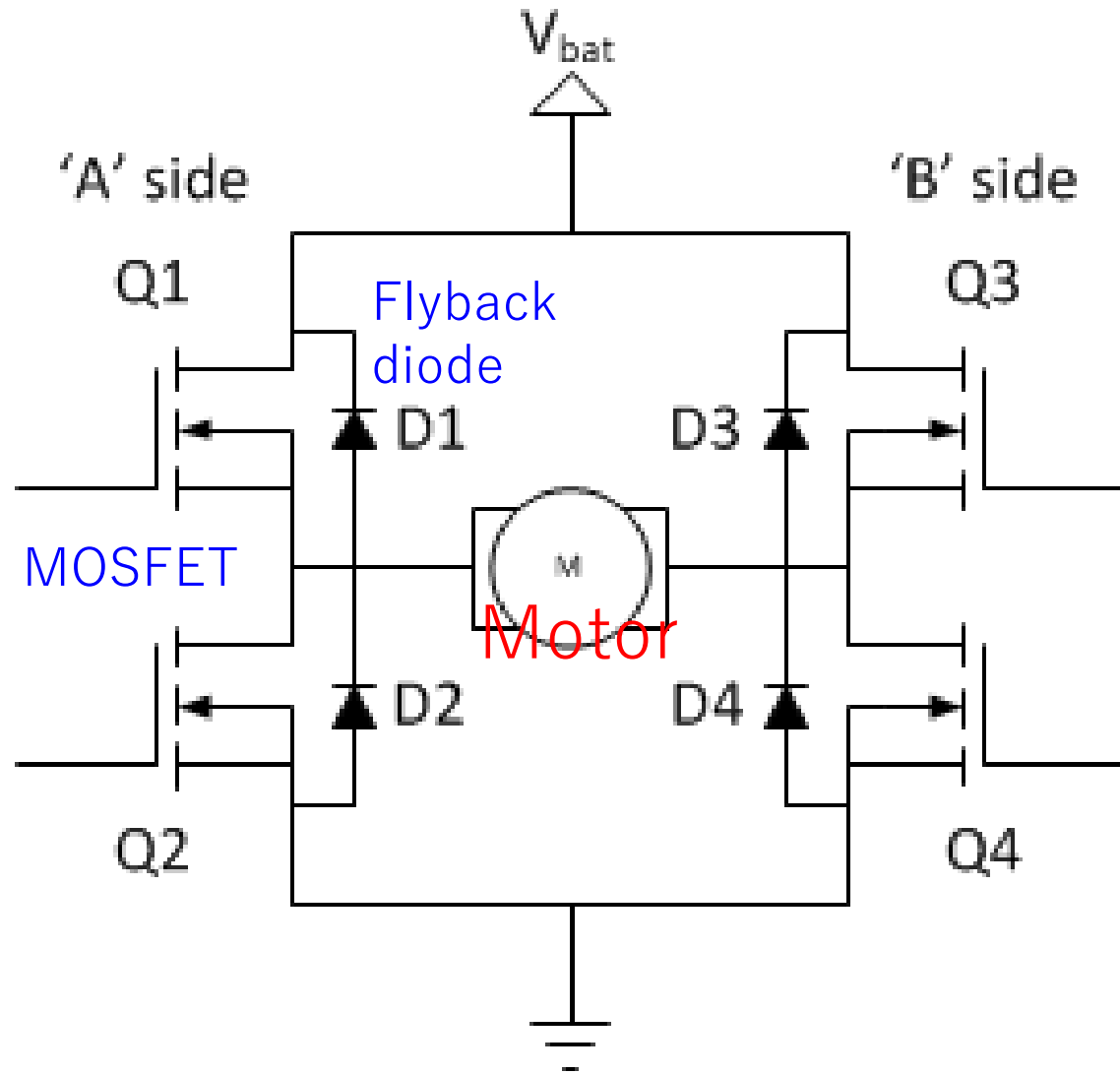
- Motor power supply voltage:
 $VM = 0-11V$
- Logic power supply voltage:
 $VCC = 2-7V$
- Continuous output current:
 $1.5A/bridge$

OCP: Over Current Protection

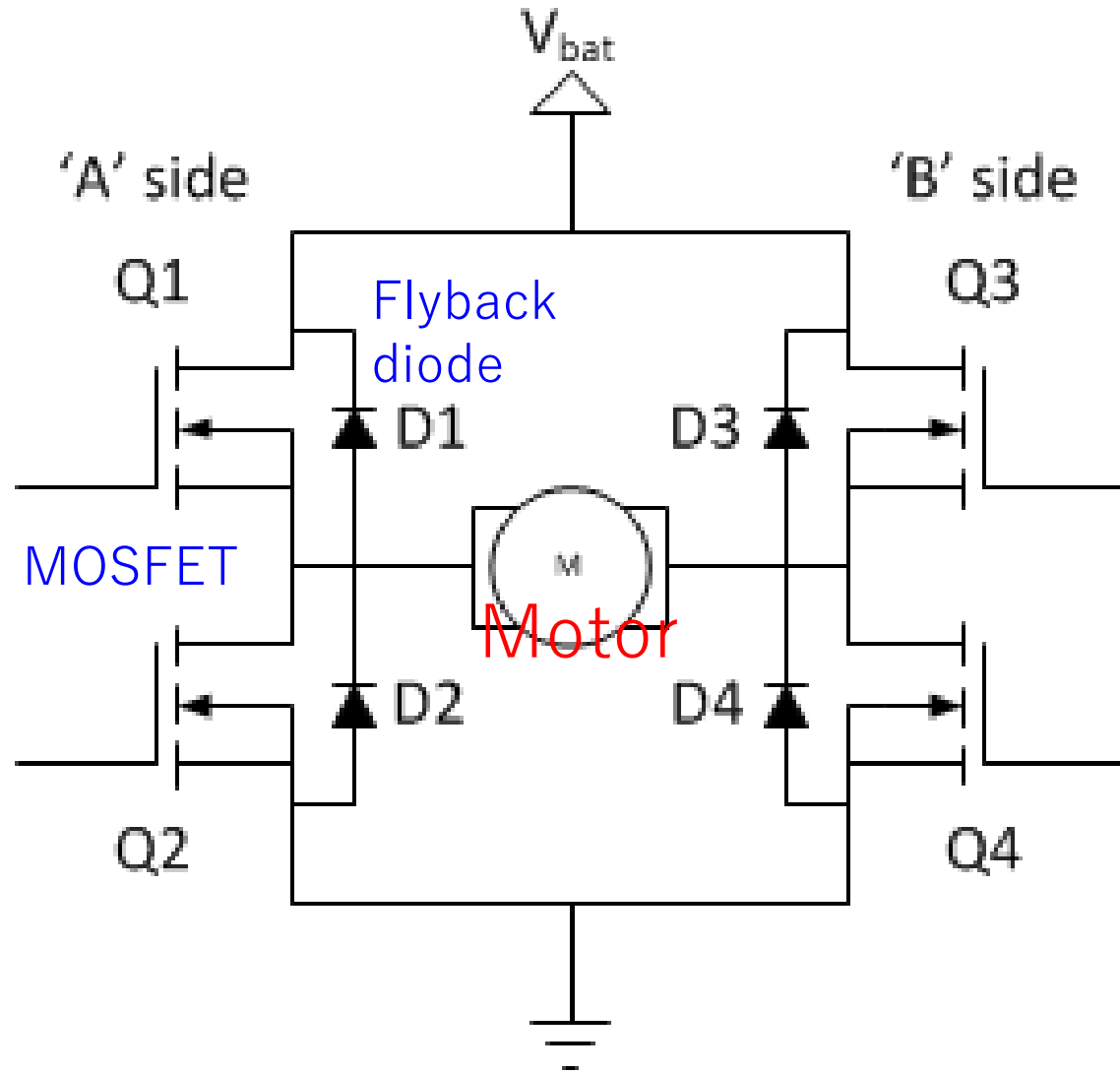
過電流保護



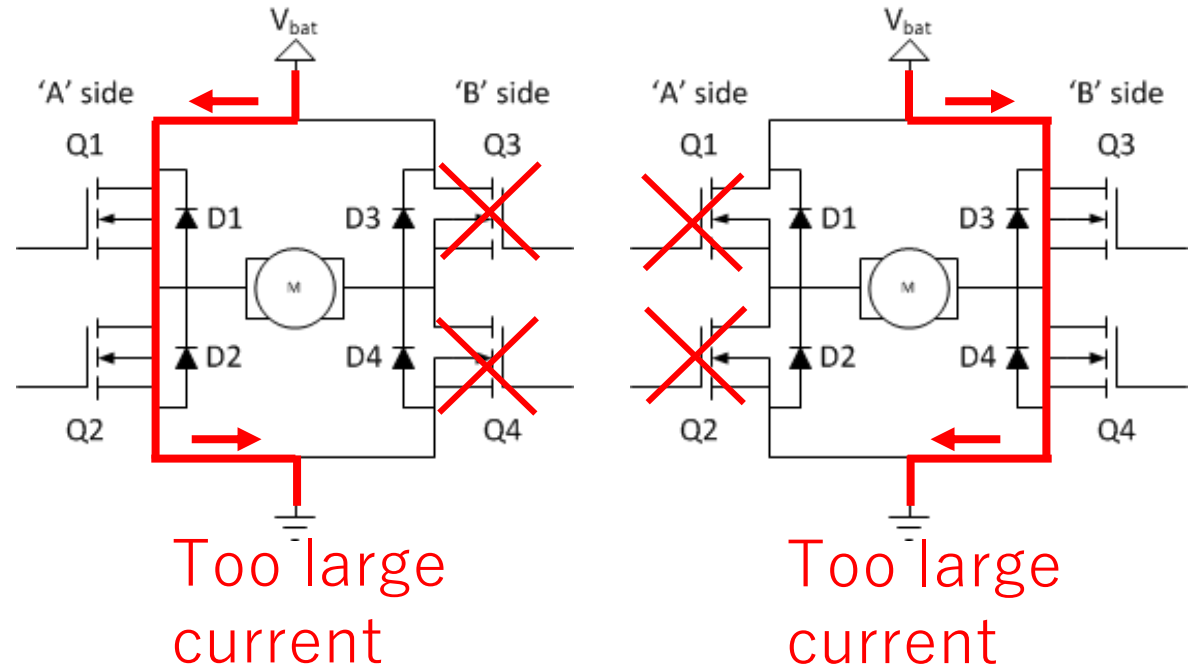
H bridge



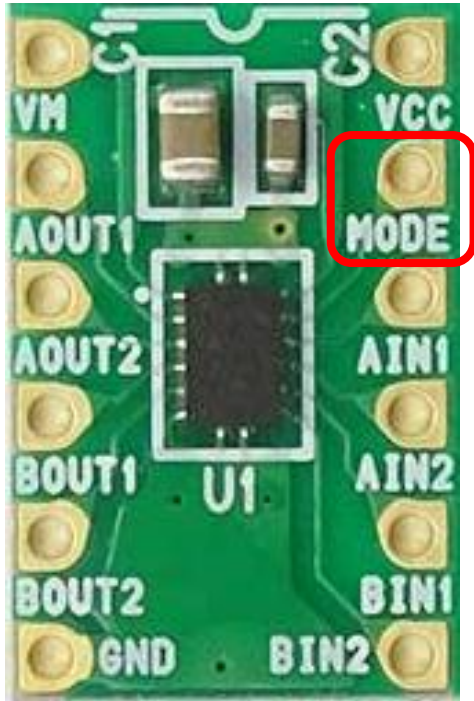
H bridge



Do not turn ON transistors on the same side



How to change rotational direction and speed



AE-DRV8835-S

- IN/IN mode (**MODE = 0**):

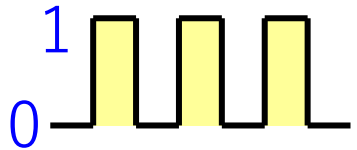
Function	Ain1	Ain2	Aout1	Aout2
Forward	1	0	H	L
Reverse	0	1	L	H
Brake	1	1	L	L
Coast	0	0	Z	Z

Z: High impedance (neither driven to a high nor low level)

- PHASE/ENABLE mode (MODE = 1):

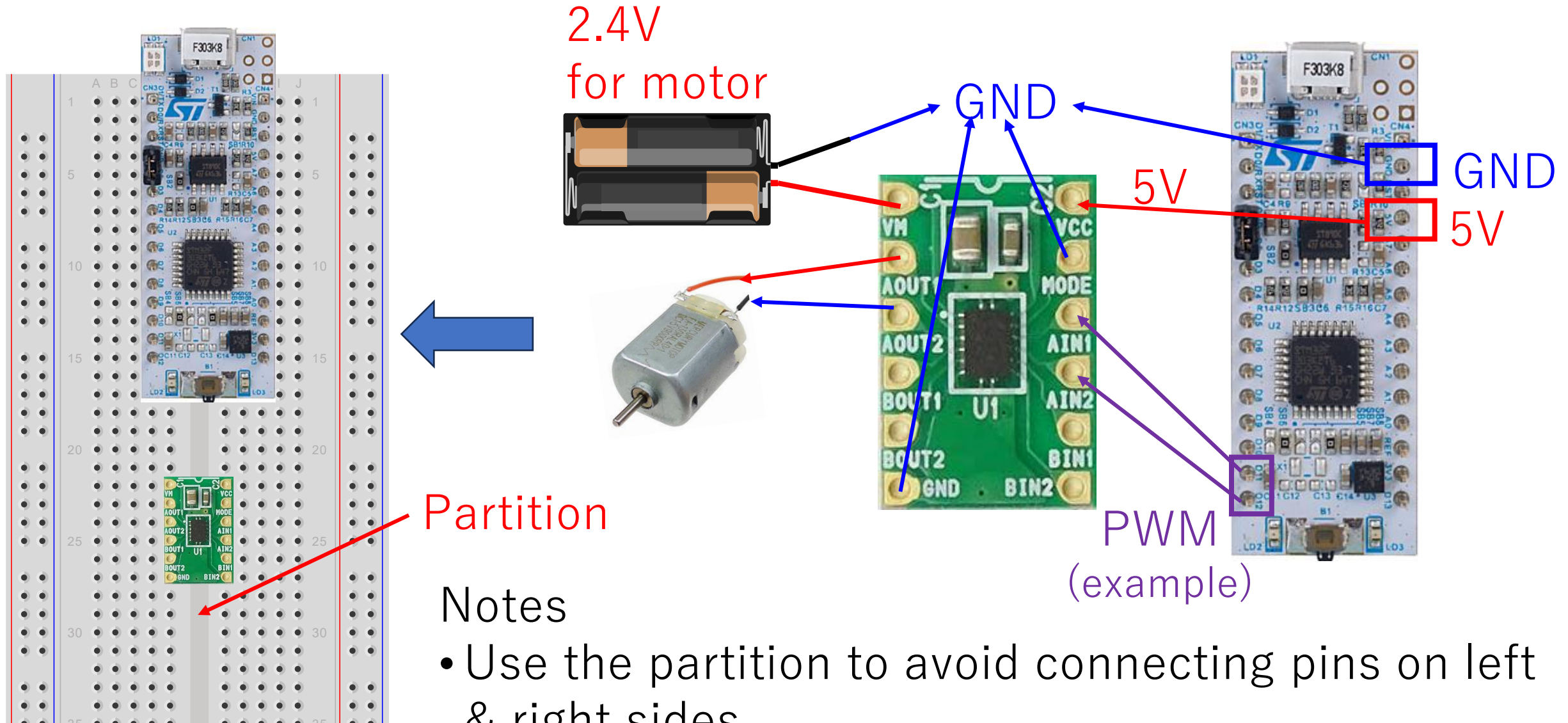
Function	Ain1	Ain2	Aout1	Aout2
Forward	1	0	H	L
Reverse	1	1	L	H
Brake	0	X	L	L

X: don't care



Change speed
using PWM (P. 11)

Step 3: Make a circuit on a breadboard



Program

2. Make a program to send a PWM signal of period 0.02s and change the rotation direction & duty cycle every 1 sec as:
Forward & 0.2 → Brake → Backward & 0.2 → Coast

- Use two “PWMOut” objects to generate two PWM signals (P. 16)
- Change “duty cycle” of two PWM signals every 1 sec as
Ain1: 0.2, Ain2: 0.0
→ Ain1: 1.0, Ain2: 1.0
→ Ain1: 0.0, Ain2: 0.2
→ Ain1: 0.0, Ain2: 0.0

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Exercise 2-6: Ultrasonic distance sensor

Exercise 2-7: Micro switch

Show the result of each exercise to an instructor or TA, every time you complete an exercise

Exercise 2-3

1. Make a circuit on a breadboard to drive a **servomotor** (SG92R)
2. Make a program to change the PWM pulse width at every 1 sec as
1.4ms $\rightarrow$ 2.2ms $\rightarrow$ 1.4ms $\rightarrow$ 0.6ms
using a PWM signal of period 0.02s
3. Then, show the result and **explain**
 - **how to compute the duty cycle from a target angle**to an instructor or a TA

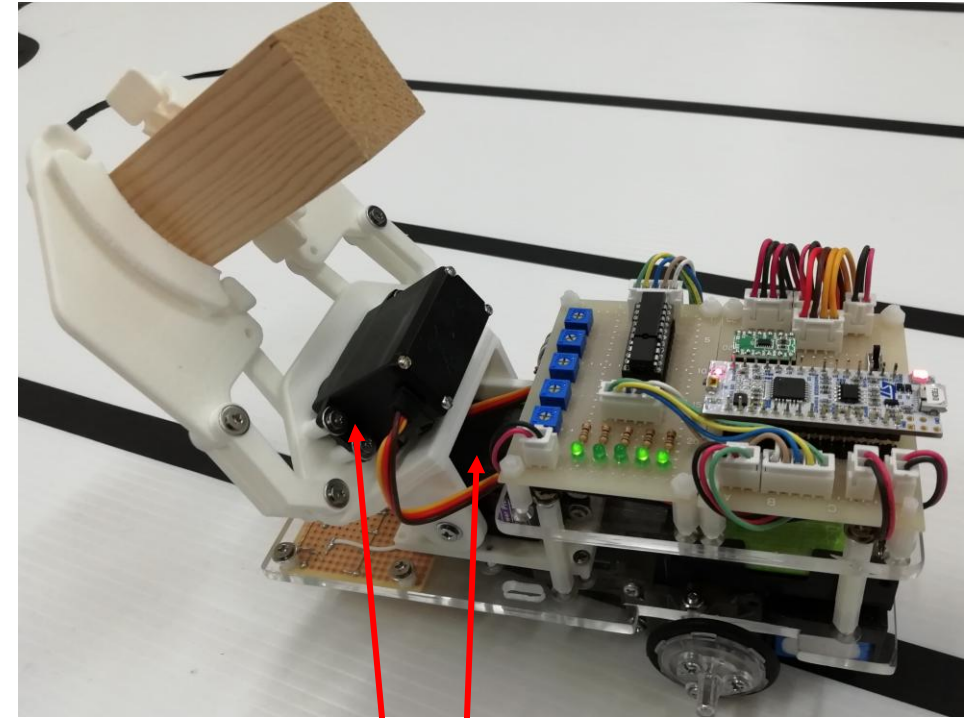
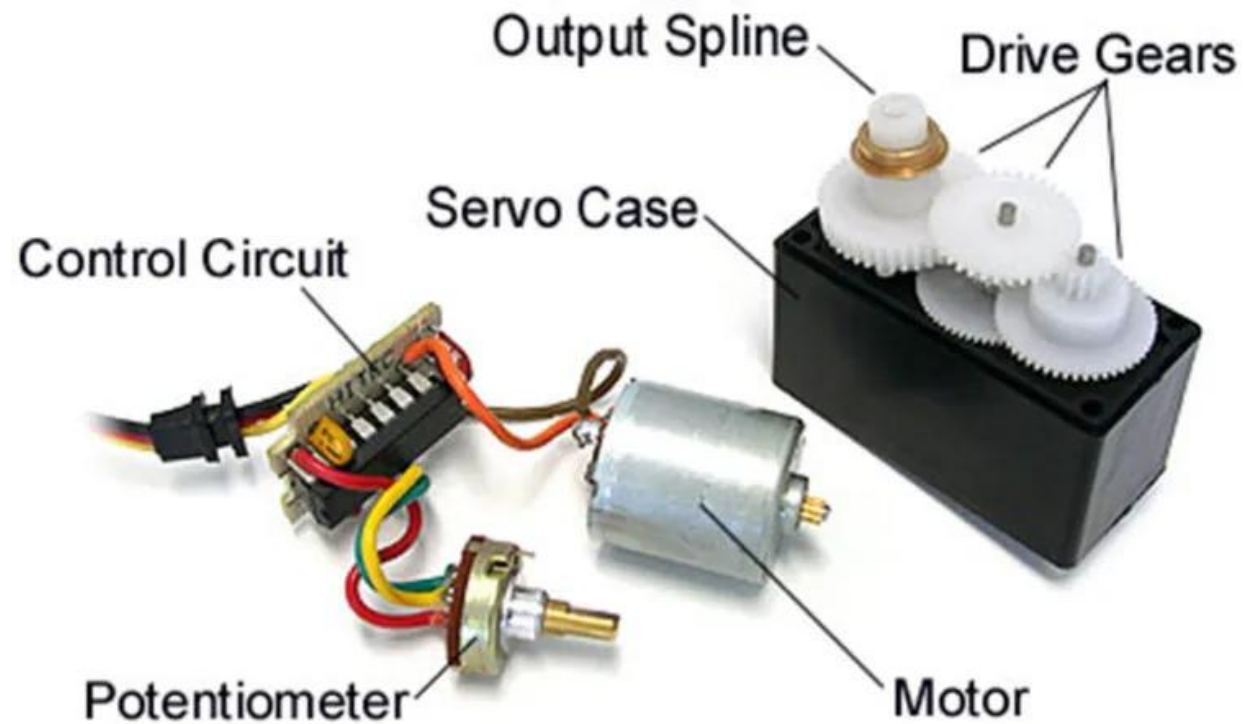
➤ See NEXT page for the procedure

Procedure

- Step 1: See P. 30-31 to learn about the servomotor
- Step 2: Choose a pin of microcontroller used for PWM (P. 12)
- Step 3: Make a circuit on a breadboard (P. 32)
- Step 4: Make a new project (P. 7-10 of “[platformIO.pdf](#)”)
- Step 5: Write a code in main.cpp (P. 33)
- Step 6: Connect a microcontroller to your PC
- Step 7: Build & run the project (P. 14 of “[platformIO.pdf](#)”)
 - See P. 16-18 of “[platformIO.pdf](#)”, if upload fails

Servomotor

Includes a DC motor, **gears**, a **sensor (potentiometer)** and **controller** to control angle



MG996R

Example of servomotor



SG92R
[Datasheet](#)



- PWM period: 20ms
- PWM pulse: 0.5ms~2.4ms
- Range : $\pm 90^\circ$
- Torque : 2.5kgf · cm
- Speed: 0.1s/60°
- Operating voltage: 4.8V
→ Use three batteries

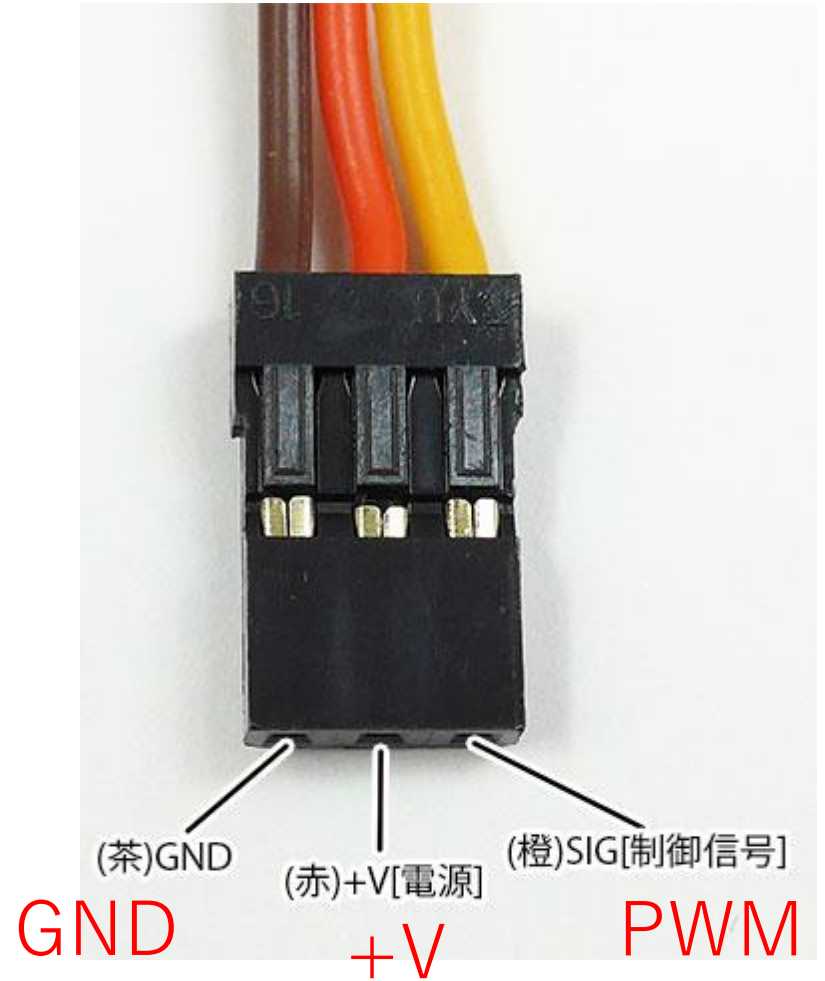
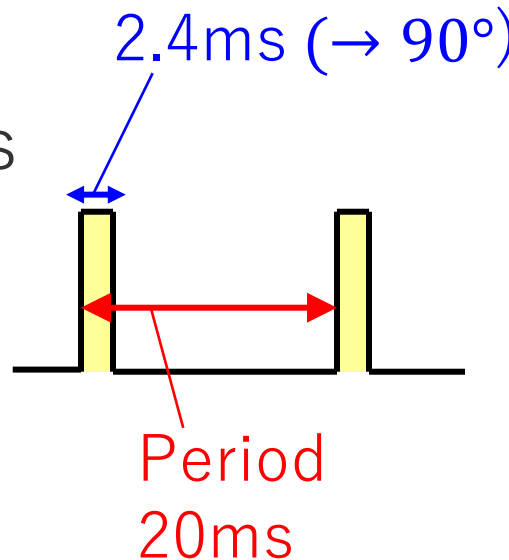
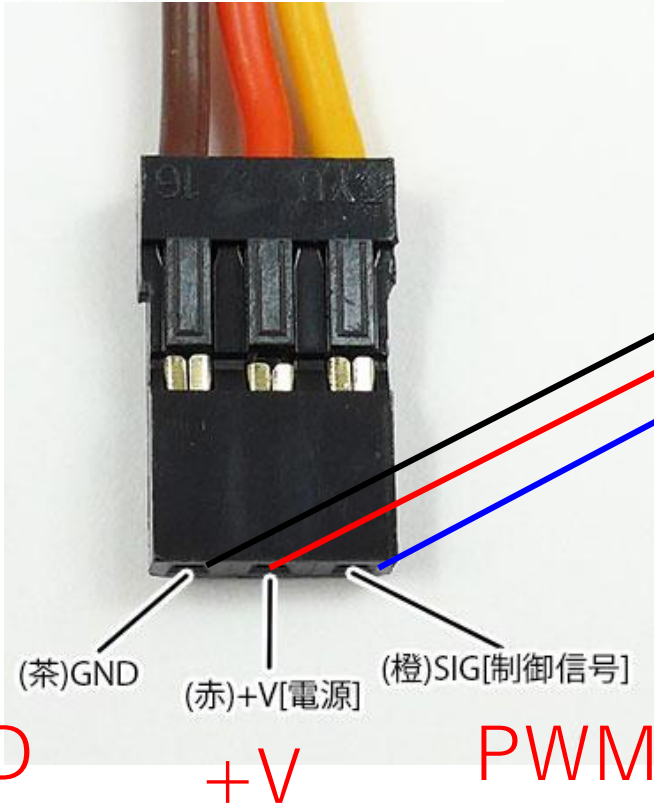


Image: <https://akizukidenshi.com/>

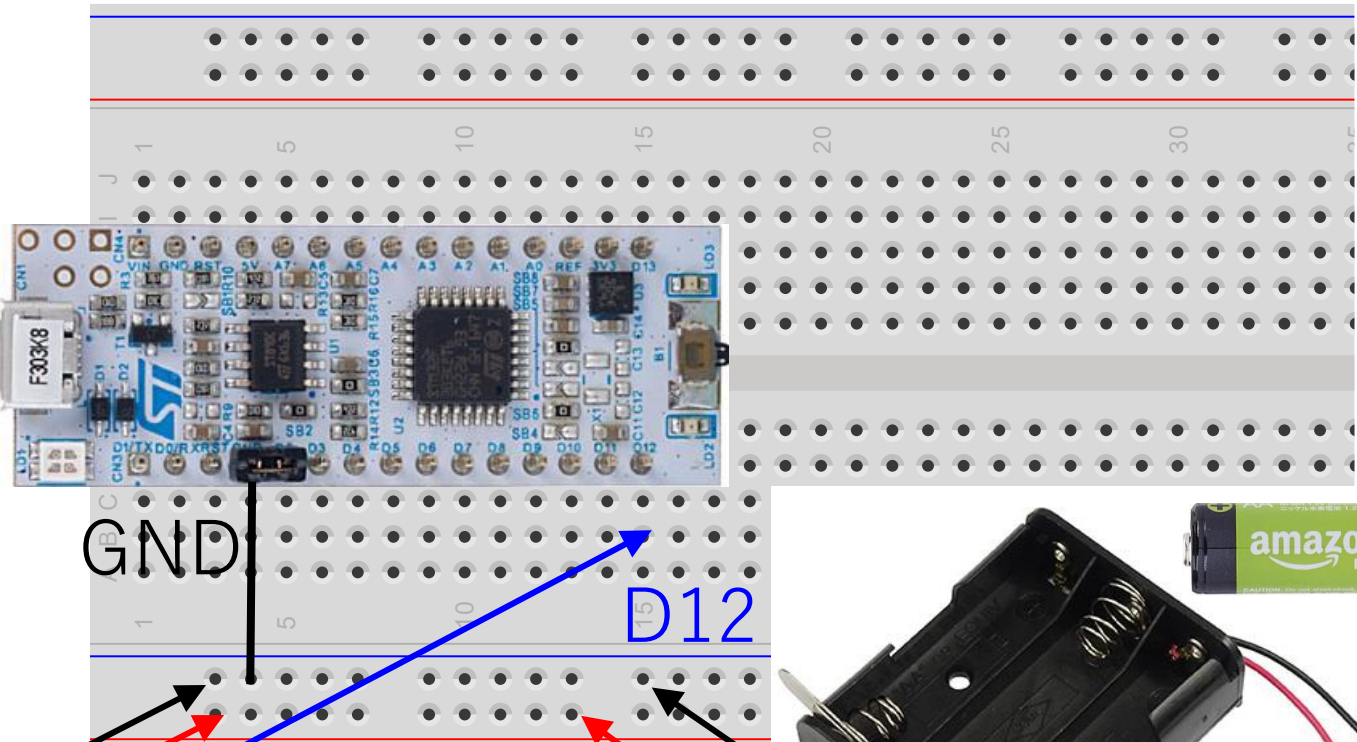
Example circuit



SG92R



GND +V PWM

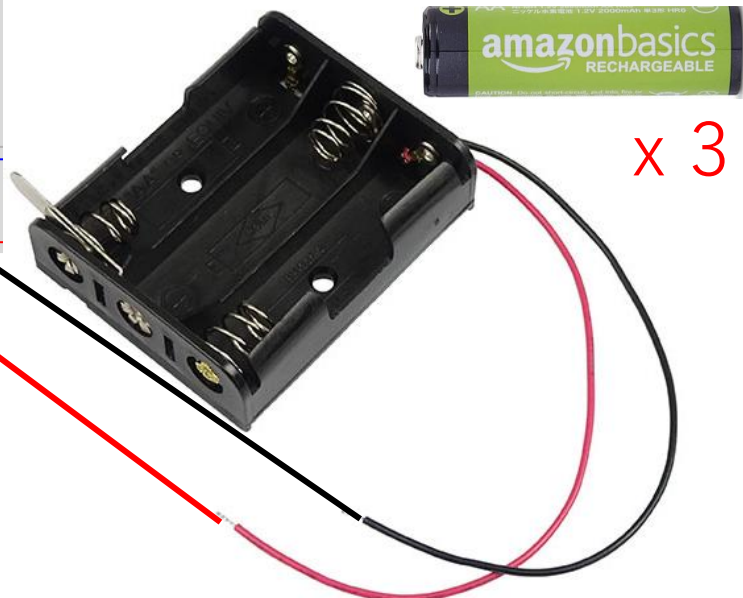


GND

D12



Jumper wires (male-male)



x 3

Program

2. Make a program to change the PWM pulse width at every 1 sec as
1.4ms $\rightarrow$ 2.2ms $\rightarrow$ 1.4ms $\rightarrow$ 0.6ms
using a PWM signal of period 0.02s.

Hint:

- Period of PWM signal is 0.02s=20ms
- Pulse width 1.4ms $\rightarrow$ Duty cycle of PWM = $1.4/20=0.07$
- Pulse width 2.2ms $\rightarrow$ Duty cycle of PWM = $2.2/20=0.11$
- Pulse width 0.6ms $\rightarrow$ Duty cycle of PWM = $0.6/20=0.03$

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Exercise 2-7: Micro switch

Show the result of each exercise to an instructor or TA, every time you complete an exercise

Exercise 2-4

- Make a program to show the distance between an object and a sensor (GP2Y0A21YK) on “Serial monitor” at every 0.1 sec.
 - Apply the program to a sensor connected to a breadboard
- See NEXT page for the procedure

Procedure

Step 1: See P. 37-38 to learn about the servomotor

Step 2: Choose a pin of microcontroller used for AnalogIn (P. 39-40)

Step 3: Make a circuit on a breadboard (P. 41)

Step 4: Make a new project (P. 7-10 of “[platformIO.pdf](#)”)

Step 5: Write a code in main.cpp (P. 42)

Step 6: Setup to **display decimal numbers** in the terminal

➤ See P. 22 of “[platformIO.pdf](#)”

Step 7: Build & run the program (P. 14 of “[platformIO.pdf](#)”)

Step 8: Check serial output in the terminal

➤ See P. 20 of “[platformIO.pdf](#)”

Infrared distance sensors

Example: GP2Y0A21YK ([Datasheet](#))

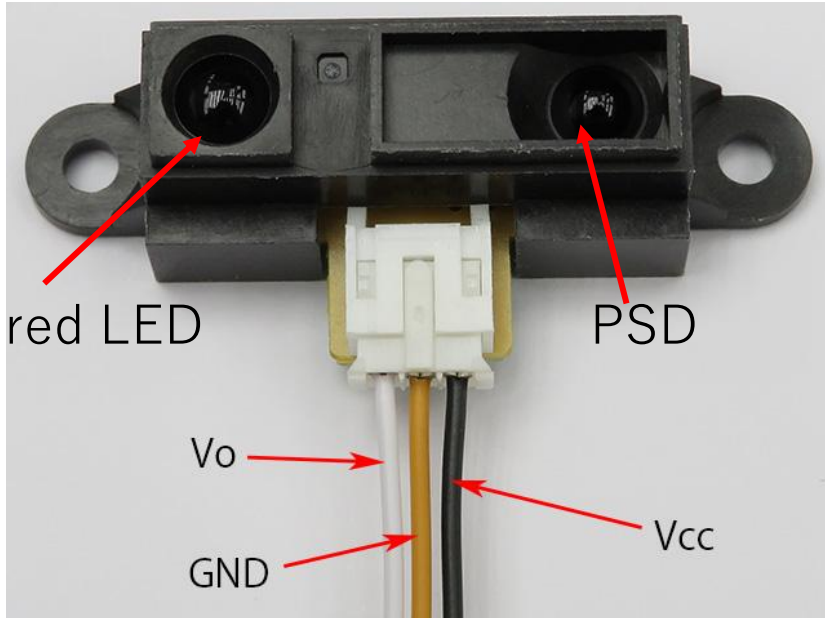


Image: <https://akizukidenshi.com/>

- Return a **voltage (V_o)** depending **on the distance** from an obstacle
- Measurable range: 10~80cm
- Supply voltage : 4.5 to 5.5V

Infrared: 赤外線

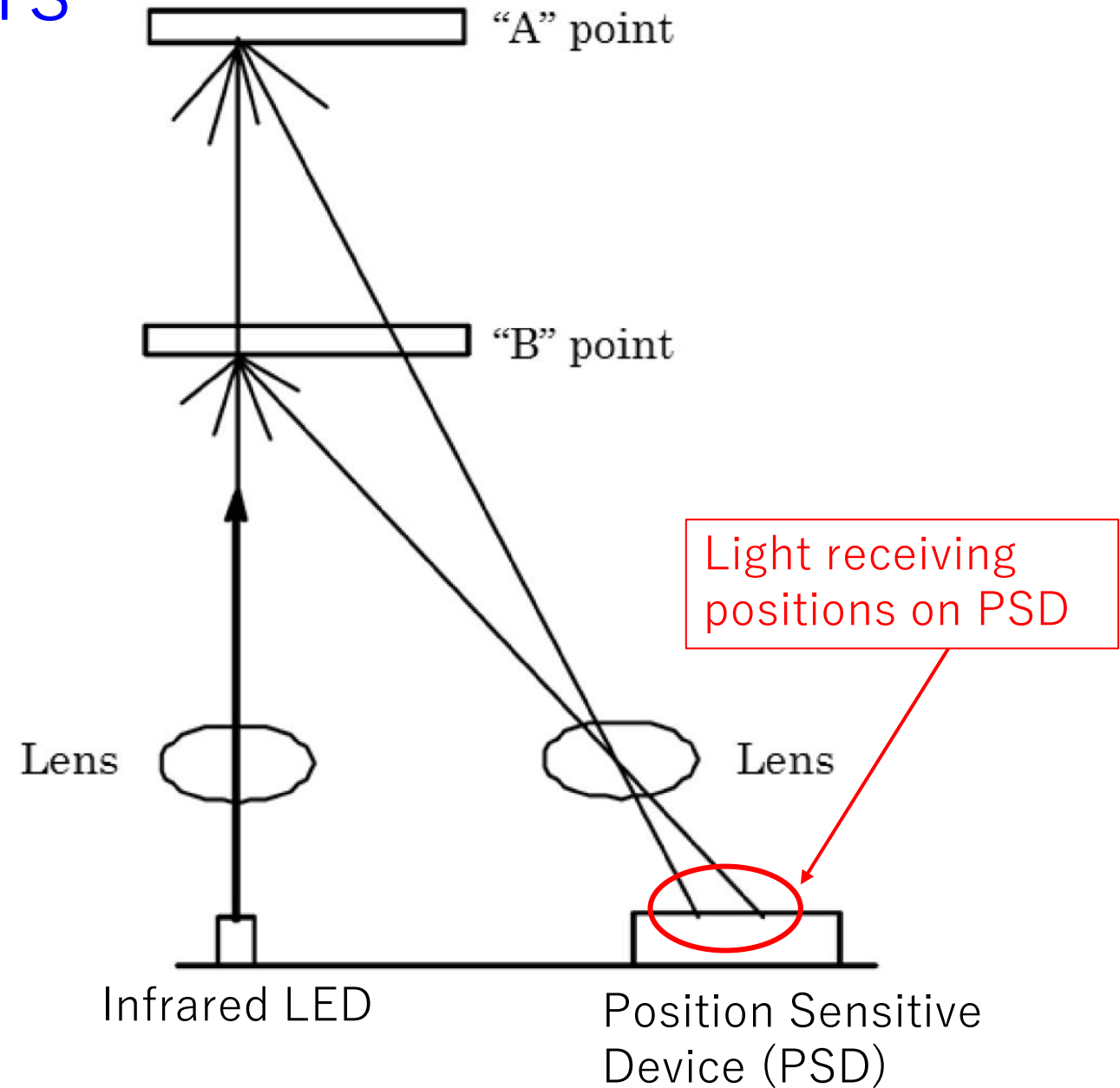
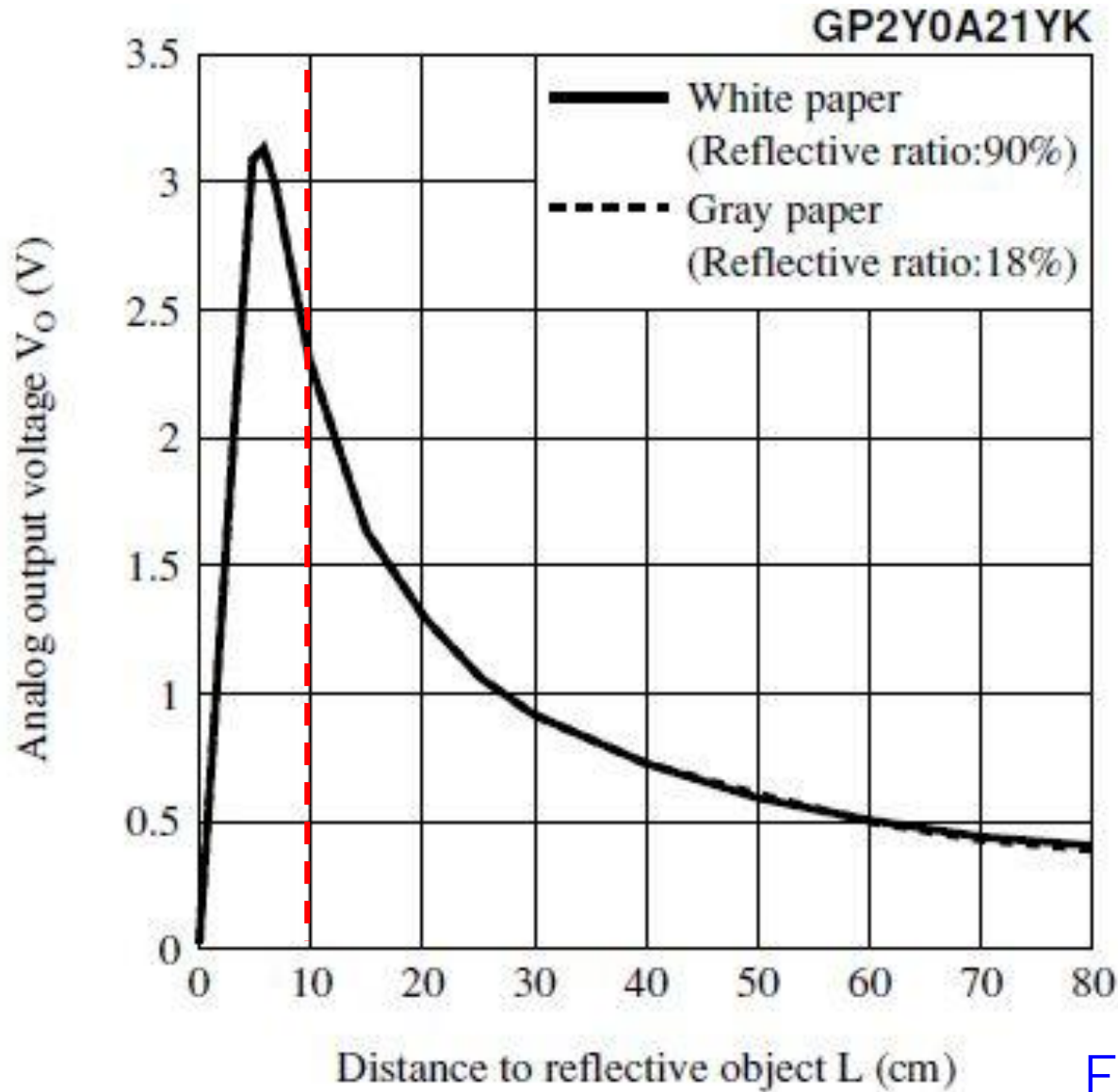


Image: <https://www.makerguides.com/>

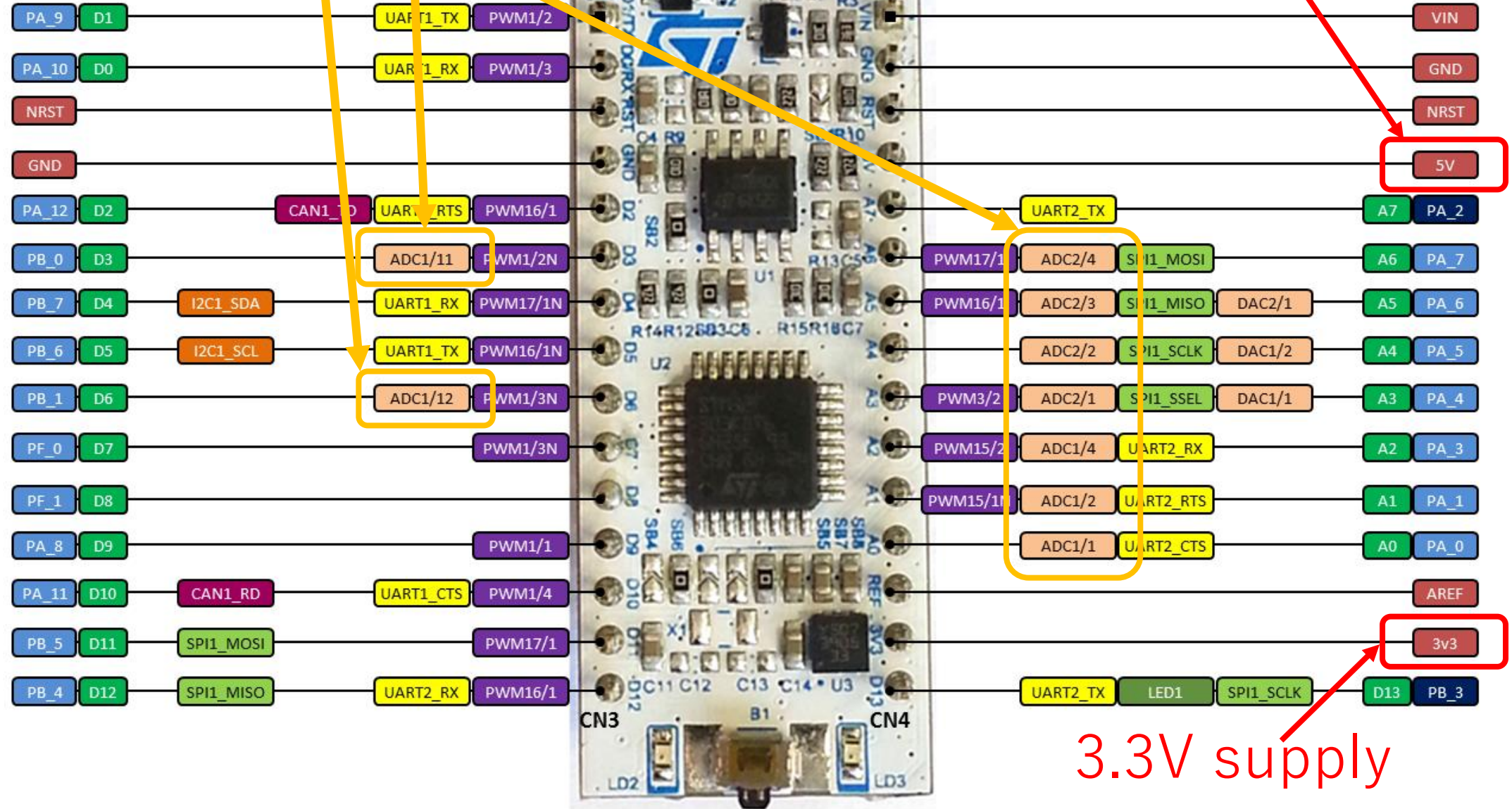
Relation between output voltage and distance



- Equations to obtain a distance from a value of V_O are estimated by several people, e.g.,
$$\text{Distance} = 26.663 * V_O^{-1.25}$$
at [this website](#).
- This equation is not correct out of the measurable range (under 10cm and over 80cm)

From [Datasheet](#)

Pins for
AnalogIN



Mbed API : “AnalogIn”

[Click here for documentation](#)

Read voltage from a pin

Initialization:

```
AnalogIn object_name(pin_name);
```

Class

Functions of AnalogIn :

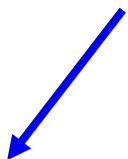
- read()
- read_voltage(),
etc.

Example:

For input voltage 1.65V

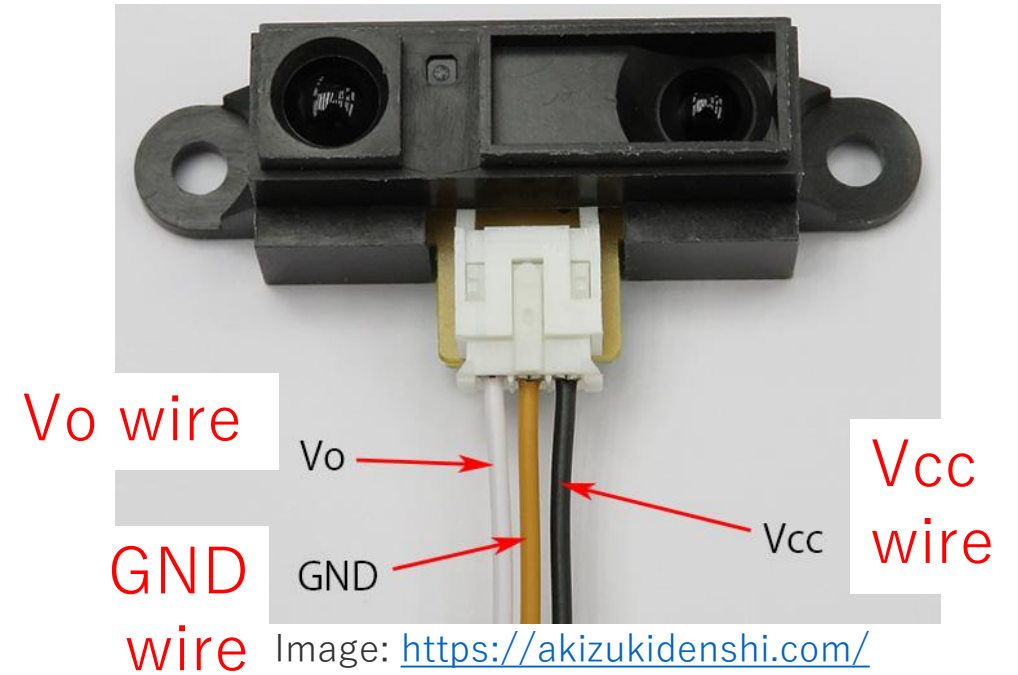
object_name.read()  0.5

$0.5 \times 3.3 \text{ V}$



Circuit

- Connect **Vo wire** to a pin for AnalogIn of microcontroller board
- Connect **GND wire** to a **GND pin** of microcontroller board
- Connect **Vcc wire** to **5V pin** of microcontroller board
(No battery is needed)



Program

- Make a program to show the distance between an object and a sensor (GP2Y0A21YK) on “Serial monitor” at every 0.1 s.
 - Use “AnalogIn” to read the voltage V_o of the sensor
 - To get the distance from an obstacle, use the equation
$$\text{Distance} = 26.663 * V_o^{-1.25}$$
 - To calculate x^y , use “pow” function as follows:
`pow(double x, double y);`
 - Use “printf” function to show the distance value
 - Use “ThisThread::sleep_for(100ms);”

Today's exercises

Exercise 2-1: Drive DC motor using transistor

Exercise 2-2: Drive DC motor using motor driver IC

Exercise 2-3: RC servomotor

Exercise 2-4: Infrared distance sensor

Exercise 2-5: Digital low-pass filter & serial plotter

Exercise 2-6: Ultrasonic distance sensor

Exercise 2-7: Micro switch

Show the result of each exercise to an instructor or TA, every time you complete an exercise

Exercise 2-5

- Modify the program in Exercise 2-4 to
 - apply the **RC filter** to the output voltage from the sensor with $a = 0.85$ and $T = 0.01$
 - calculate the distances using **both filtered and unfiltered voltages**
 - Display the **time responses** of **both filtered and unfiltered distances** on “**Teleplot**”.
- See NEXT page for the procedure

Procedure

Step 1: See P. 46-47 to learn about the RC filter

Step 2: Choose a pin of microcontroller used for AnalogIn
(P. 39)

Step 3: Use the same circuit for Exercise 2-4

Step 4: Install “Teleplot”.

➤ See P. 24-25 of “[platformIO.pdf](#)”

Step 5: Modify the code in main.cpp (P. 48)

Step 6: Build & run the program (P. 14 of “[platformIO.pdf](#)”)

Step 7: Check graphs in “Teleplot”.

➤ See P. 27-29 of “[platformIO.pdf](#)”

Digital low-pass filter

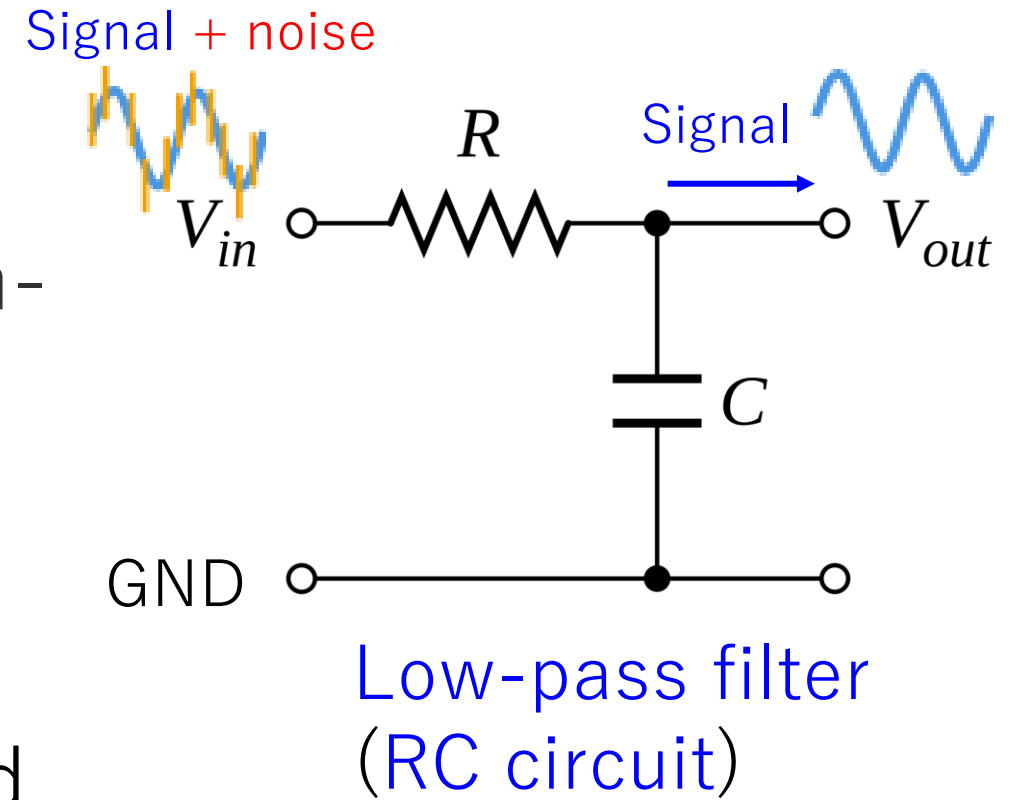
Example: RC filter

- Smooth signals by removing high-frequency components
- Cutoff frequency:

$$f_c = \frac{1}{2\pi RC} \text{ [Hz]}$$

(boundary between a passed and removed frequencies)

- Achieve the function of RC circuits by **software**



V_{in} : Filter input voltage
(signal from sensor)
 V_{out} : Filter output voltage

Digital low-pass filter

Example: RC filter

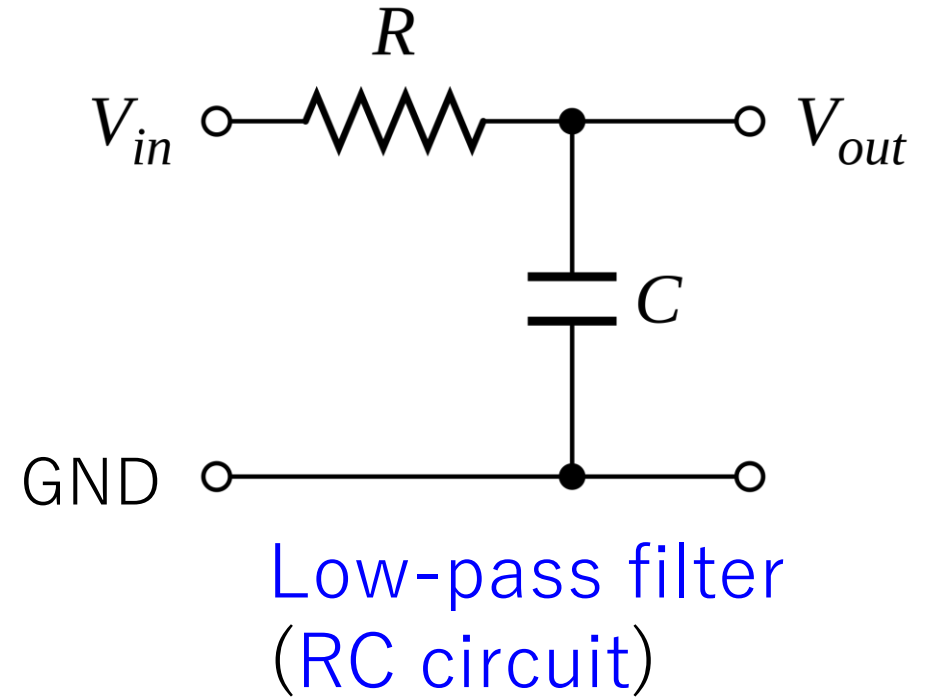
Calculation of V_{out} at each time step
(every T seconds)

$$V_{out} = a \times V_{out}^{pre} + (1 - a) \times V_{in}$$

$$a = \frac{1}{2\pi f_c T}$$

f_c : cutoff frequency

V_{out}^{pre} : previous value of V_{out}
(the value T seconds before)



V_{in} : Filter input voltage
(signal from sensor)
 V_{out} : Filter output voltage

Program

- Modify the program in Exercise 2-4 to
 - apply the **RC filter** to the output voltage from the sensor with $a = 0.85$ and $T = 0.01$
 - calculate the distances using **both filtered and unfiltered voltages**

Hint:

- Use $V_f = a \times V_f + (1 - a) \times V_o$ with $a = 0.85$ and the initial value 0.0 of V_{out} to get filtered voltage
- To get the distance from an obstacle, use the equation
$$\text{Distance} = 26.663 * V_f^{-1.25}$$
- Use “printf” function to show the distance (P. 27 of “**platformIO.pdf**”)
- Use “`ThisThread::sleep_for(10ms);`”

Today's exercises

Exercise 2-1: Drive DC motor using transistor

Exercise 2-2: Drive DC motor using motor driver IC

Exercise 2-3: RC servomotor

Exercise 2-4: Infrared distance sensor

Exercise 2-5: Digital low-pass filter & serial plotter

Exercise 2-6: Ultrasonic distance sensor

Exercise 2-7: Micro switch

Show the result of each exercise to an instructor or TA, every time you complete an exercise

Exercise 2-6

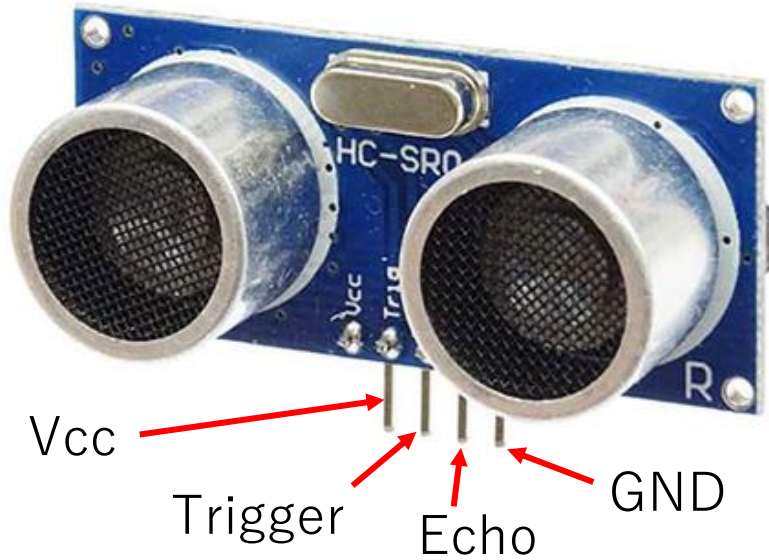
- Make a program to show the distance between an object and a sensor (HC-SR04) on “Serial monitor”
- Apply the program to a sensor connected to a breadboard

➤ See NEXT page for the procedure

Procedure

- Step 1: See P. 52 to learn about the sensor
- Step 2: Choose pins (D0-D13, A0-A7) of microcontroller to connect “trigger” and “echo” pins
- Step 3: Make a circuit on a breadboard (P. 53)
- Step 4: Make a new project (P. 7-10 of “[platformIO.pdf](#)”)
- Step 5: Add a library (P. 31-33 of “[platformIO.pdf](#)”)
- Step 6: Copy an example code (P. 34 of “[platformIO.pdf](#)”)
- Step 7: Build & run the program (P. 14 of “[platformIO.pdf](#)”)
- Step 8: Check serial output in the terminal
 - See P. 20 of “[platformIO.pdf](#)”

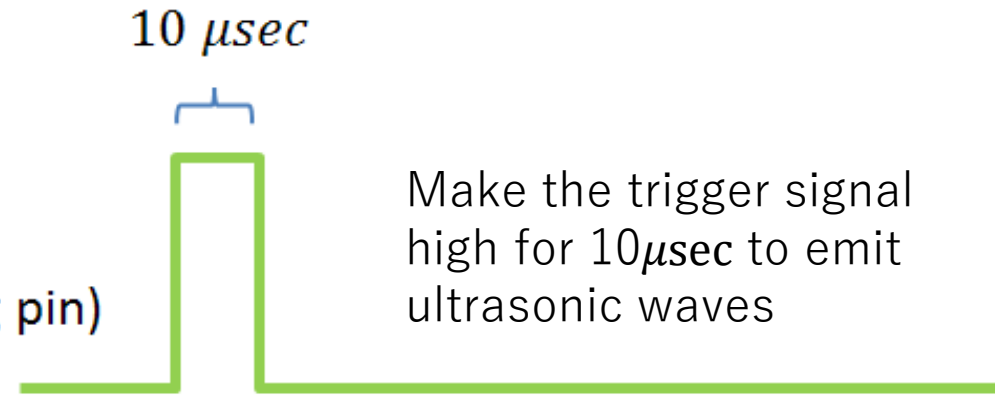
Ultrasonic distance sensors



HC-SR04 [Click for datasheet](#)

- Distance measuring range :
2 cm to 4 m
- Supply voltage : 5.0V

Trigger signal
(TTL input to Trig pin)



Echo signal
(TTL output from Echo pin)

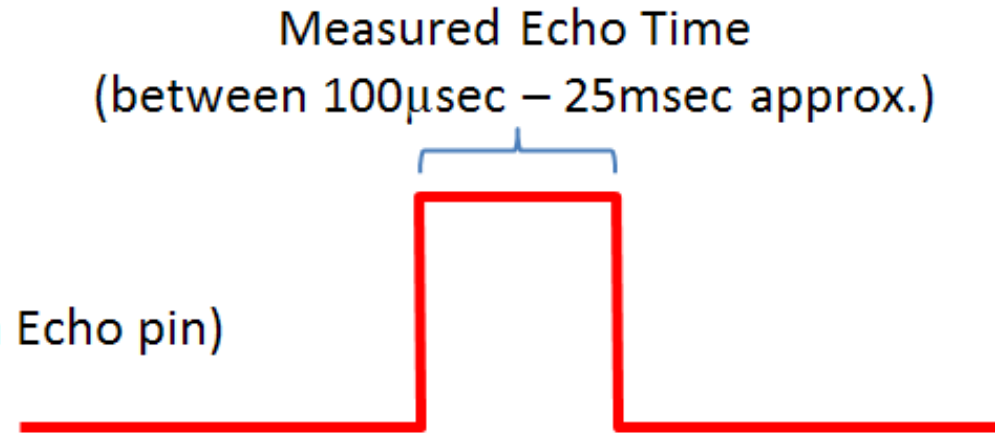


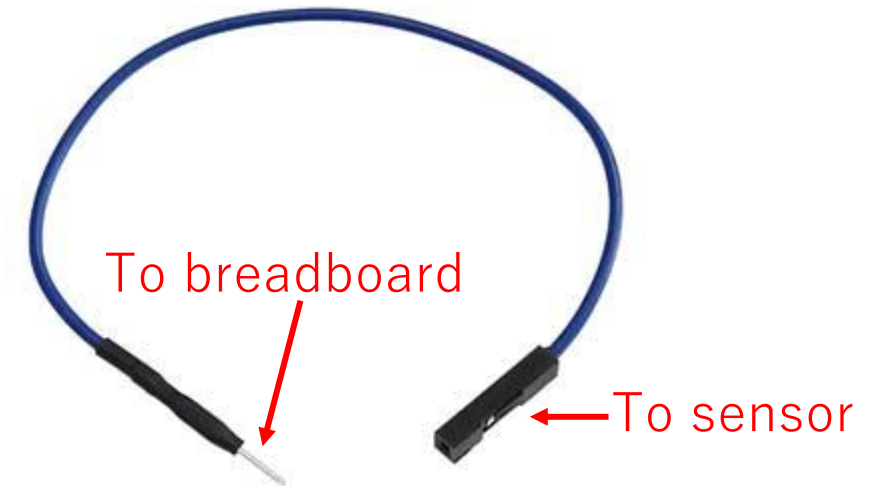
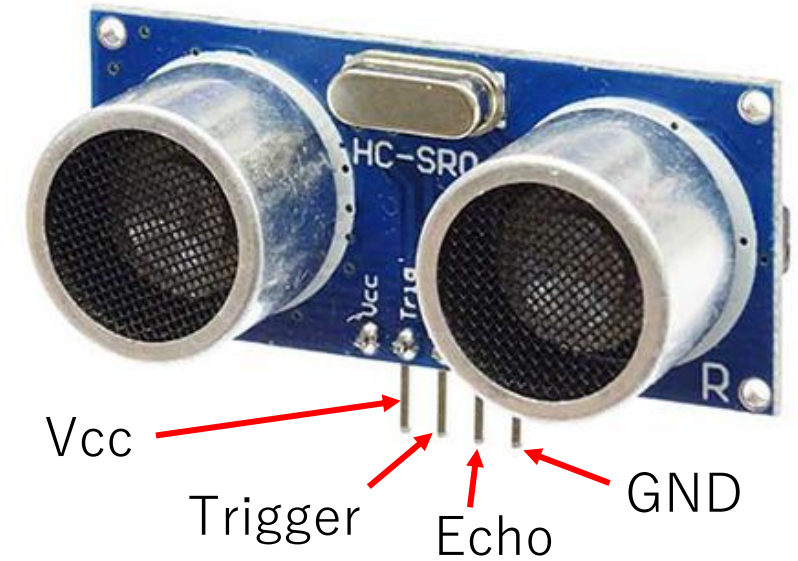
Image: <https://os.mbed.com/>

$$Dist[cm] = \frac{Echo\ time[\mu sec] \times Sonic\ speed[cm/\mu sec]}{2}$$

$$Sonic\ speed = 0.0343[cm/\mu sec]$$

Circuit

- Use “female-male” jumper wires to connect a sensor to a breadboard
- Connect Vcc pin to 5V pin of microcontroller board
- Connect GND pin to a GND pin of microcontroller board
- Connect “trigger” and “echo” pins to microcontroller pins (e.g., D0-D13, A0-A6)



“Female-male” jumper wire

Today's exercises

Exercise 2-1: Drive DC motor using transistor

Exercise 2-2: Drive DC motor using motor driver IC

Exercise 2-3: RC servomotor

Exercise 2-4: Infrared distance sensor

Exercise 2-5: Digital low-pass filter & serial plotter

Exercise 2-6: Ultrasonic distance sensor

Exercise 2-7: Micro switch

Show the result of each exercise to an instructor or TA, every time you complete an exercise

Exercise 2-7

1. Make a circuit on a breadboard using a micro switch (SS-10GL13) and a program to display “low” if SS-10GL13 is pushed and display “high” otherwise.
2. Then, show the difference results between
 - when the pull-up switch is NOT activated and
 - when the pull-up switch is activatedto an instructor or a TA

➤ See NEXT slide for the procedure

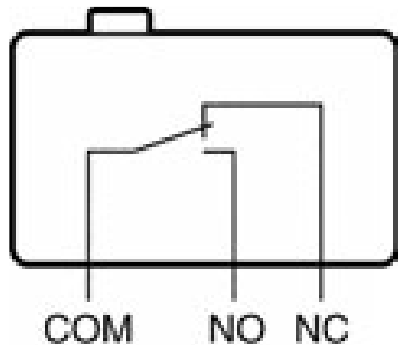
Procedure

- Step 1: See P. 57-58 to learn about the micro switch and a pull-up resistor for a switch
- Step 2: Choose a pin (D0-D13, A0-A7) of microcontroller to connect “COM” pin
- Step 3: Make a circuit on a breadboard (P. 59)
- Step 4: Make a new project (P. 7-10 of “[platformIO.pdf](#)”)
- Step 5: Write a code in main.cpp (P. 61-62)
- Step 6: Build & run the program (P. 14 of “[platformIO.pdf](#)”)
- Step 7: Check serial output in the terminal
 - See P. 20 of “[platformIO.pdf](#)”

Micro switches



COM: Common
NO : Normally Open
NC: Normally Close



Ultrasonic sensor

Micro switches

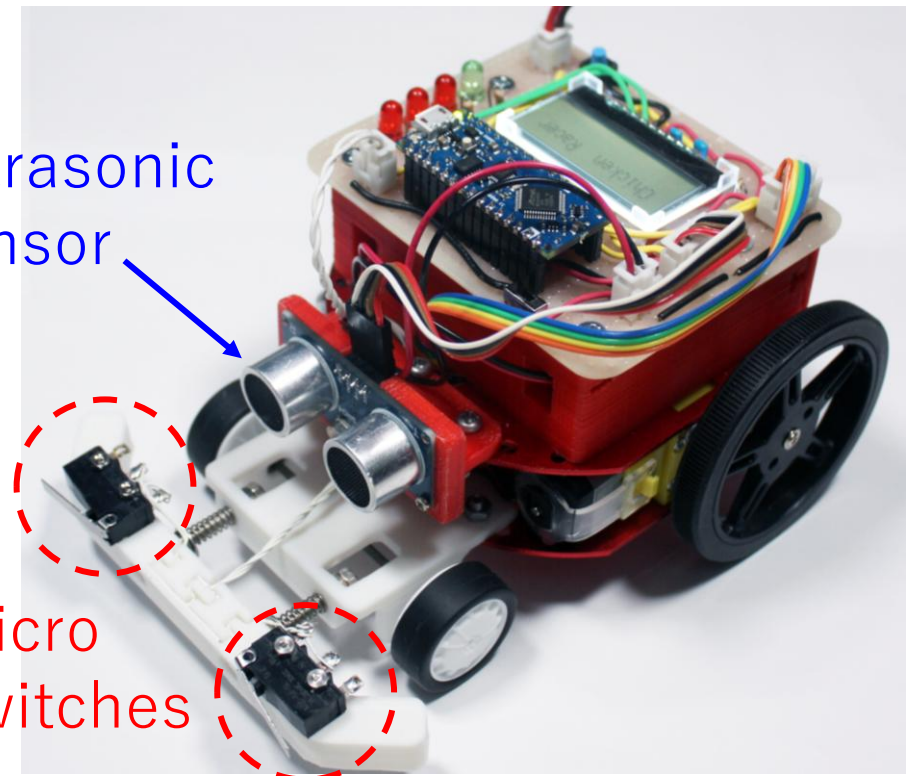
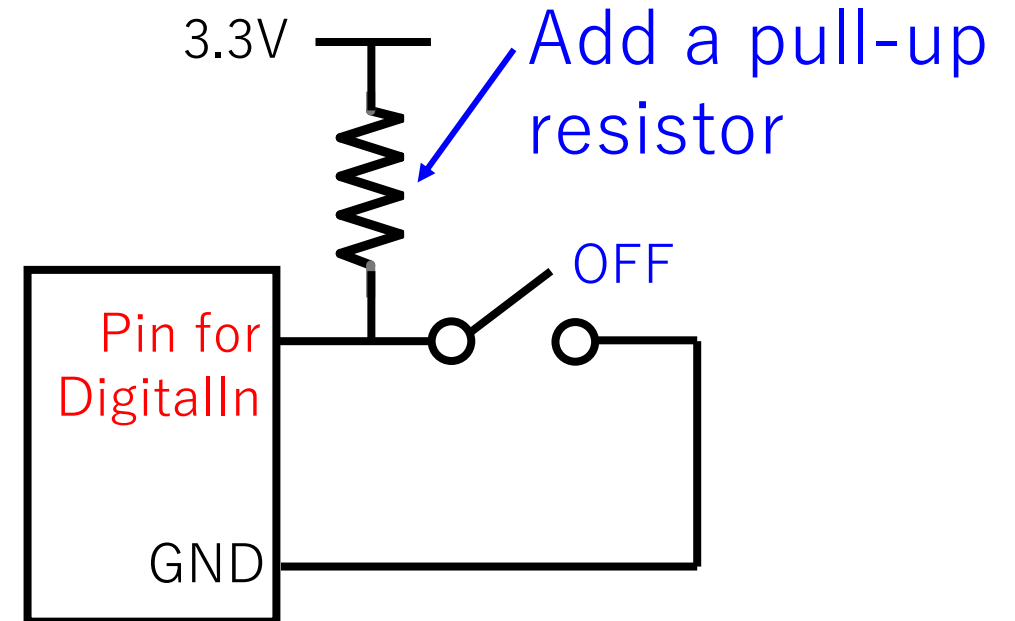
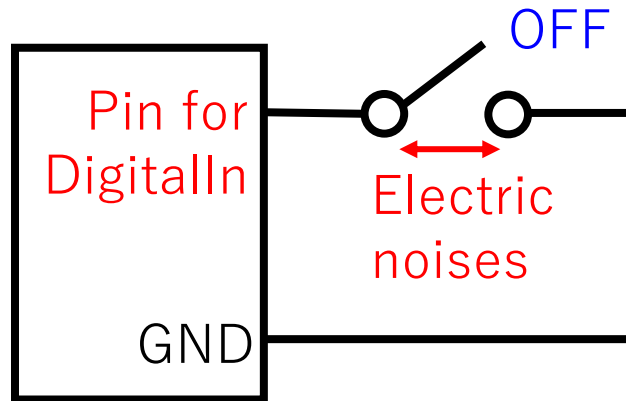


Image source: <https://deviceplus.jp/>

- If an object push the lever, the button will be pushed
- Then, “NO” and “COM” will be connected
- Different types of levers, terminals, etc. are available

Pull-up resistor for switch



Not clear if a pin is High or Low when a switch is OFF

A pin is High when a switch is OFF

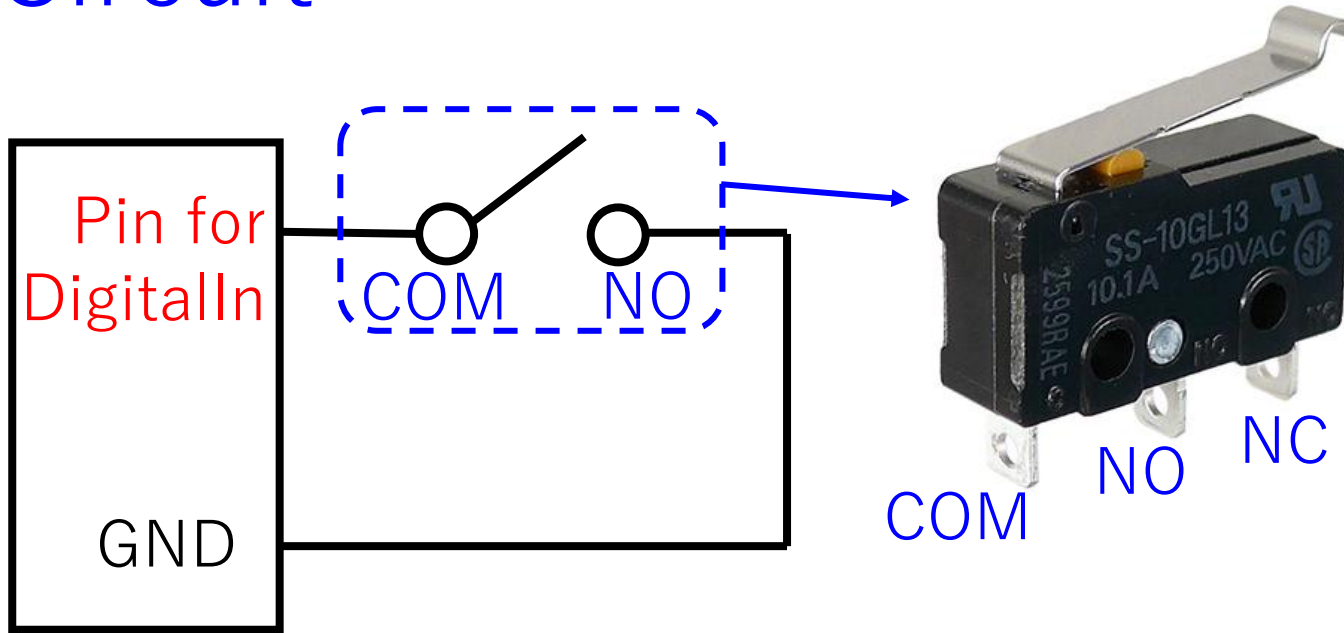
General problem of switches



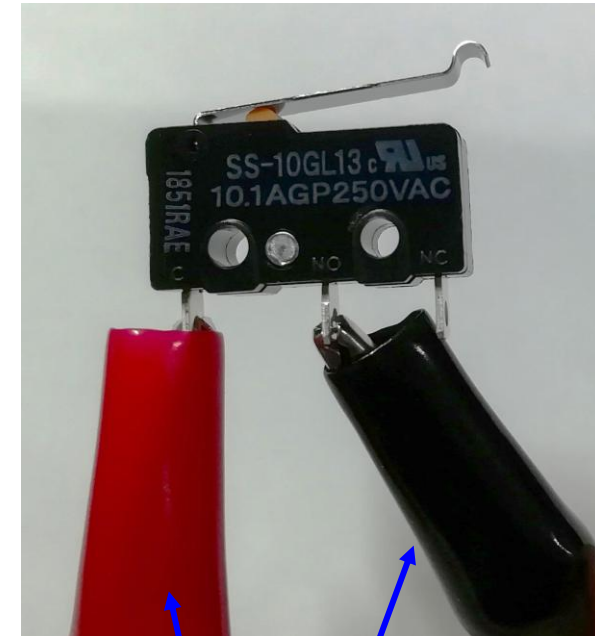
NOTE:

NUCLEO-F303K8 has pull-up resistors inside
→ No need to add a resistor in this exercise

Circuit



- Use two alligator clips in a box to connect a microswitch to a breadboard (NO→GND, COM→Pin for DigitalIN)
- NC is not connected



Alligator clips in a box

Program

- Make a circuit on a breadboard using a micro switch (SS-10GL13) and a program to display “low” if SS-10GL13 is pushed and display “high” otherwise.

- Initialize `DigitalIn` (P. 61) object as `DigitalIn object_name(Pin);`
- Use `.read()` function to read a digital input value from a pin
- Use `.mode(PullUp)` to activate the **pull-up resister**
- Use if statement to print “high” or “low” depending on a digital input value
- Use `“ThisThread::sleep_for();”`

Mbed API : “DigitalIn”

Read a digital input value from a pin

Initialization:

```
DigitalIn object_name(Pin);
```

Member functions:

- `read()`: read the value of *Pin*
- `mode()`: activate the **pull-up resistor**, pull-down resistor, etc.

[Click for Documentation](#)

Example:

```
DigitalIn din(D12);

int main(){
    din.mode(PullUp);
    while(1){
        if(din.read()==1){
            ...
        }
        ...
    }
}
```